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Determinants of Employment Across Europe: Human Capital, Gender, Family, Immigration, and Institutional Effects

Contents

1. INTRODUCTION	4
1.1 Background and Motivation.....	4
1.2 Research Problem and Significance.....	5
1.3 Research Objectives	6
1.4 Research Questions	7
1.5 Research Hypotheses	8
1.6 Scope and Limitations.....	11
1.7 Structure of the Thesis	12
Chapter 2: Theoretical Framework	14
2.1 Introduction to Theoretical Foundations	14
2.2 Human Capital Theory.....	15
2.2.1 Foundations (Becker, Mincer, Schultz)	15
2.2.2 Education and Employment Nexus	16
2.2.3 Empirical Predictions	17
2.3 Labour Supply Theory	19
2.3.1 Labour–Leisure Choice Model	19
2.3.2 Reservation Wage Theory	20
2.3.3 Application to Employment Decisions.....	21
2.4 Discrimination Theories	23
2.4.1 Taste Based Discrimination (Becker)	23
2.4.2 Statistical Discrimination (Arrow, Phelps)	24
2.4.3 Gender Specific Mechanisms	25
2.5 Immigration Economics	26
2.5.1 Immigrant Labour Market Integration	26
2.5.2 Host Country Specific Human Capital	27
2.6 Institutional Theory	29
2.6.1 Welfare Regime Typology (Esping Andersen)	29
2.6.2 Institutional Effects on Employment	30
2.7 Synthesis: Integrated Theoretical Model.....	32
2.8 Chapter Summary and Link to Empirical Literature	33
3. REVIEW OF EMPIRICAL RESEARCH.....	35
3.1 Introduction and Review Methodology	35

3.2 Education and Human Capital Effects	36
3.2.1 Cross Country Evidence	36
3.2.2 Magnitude of Education Effects	38
3.2.3 Methodological Approaches	39
3.2.4 Research Gaps	41
3.3 Gender and Employment.....	43
3.3.1 Gender Gap Evidence Across Europe	43
3.3.2 Mechanisms and Explanations	44
3.3.3 Policy Interventions	46
3.4 Family Structure and Employment.....	48
3.4.1 Marriage Effects	48
3.4.2 Motherhood/Fatherhood Penalties/Premiums	49
3.4.3 Cross National Variation	50
3.5 Immigration and Employment	52
3.5.1 Immigrant Employment Gaps	52
3.5.2 Integration Patterns	54
3.5.3 Policy Context.....	55
3.6 Health, Disability and Employment.....	57
3.6.1 Health–Employment Relationship	57
3.6.2 Disability Policy Effects	58
3.7 Institutional and Cross Country Studies	59
3.7.1 Comparative Employment Research	59
3.7.2 Welfare Regime Effects	61
3.8 Synthesis of Literature	62
3.9 Research Gaps This Thesis Addresses	65
3.10 Chapter Summary and Transition to Methodology	66
Plan	68
References	75

1. INTRODUCTION

1.1 Background and Motivation

This section situates the study within the broader context of European labour markets and explains why employment outcomes cannot be understood through individual characteristics alone. It argues that employment is the result of an interaction between personal attributes and institutional environments, an interaction that varies substantially across countries and over time.

Employment outcomes differ markedly across Europe even among individuals with similar observable characteristics. Workers with comparable levels of education, age, and work experience often face very different employment prospects depending on the country in which they reside. For example, gender based differences in employment probabilities remain limited in countries such as Sweden, while they are substantially larger in Southern European countries, particularly among parents of young children. These contrasts cannot be explained by education or skills alone, indicating the importance of national institutional contexts.

A large body of economic research emphasizes the role of human capital in shaping labour market outcomes. Human capital theory predicts that higher levels of education increase productivity and, consequently, employment opportunities (Becker, 1964; Mincer, 1974). Empirical evidence across developed economies consistently supports this relationship. However, education does not operate in isolation. The transferability of skills, recognition of qualifications, and opportunities to utilize education vary across institutional settings. Immigrants may experience difficulties converting educational attainment into employment when foreign credentials are undervalued or language barriers persist (Chiswick & Miller, 2009). Similarly, individuals with health limitations or disabilities may face structural barriers that weaken the expected returns to education.

Gender and family structure further complicate the relationship between individual characteristics and employment. Although female labour force participation has increased substantially across Europe, large cross-country differences remain (Blau & Kahn, 2013). The presence of children affects employment decisions differently across welfare regimes, reflecting variation in childcare provision, parental leave policies, and cultural norms regarding caregiving. In some institutional contexts, family responsibilities significantly reduce women's employment probabilities, while in others their effects are minimal.

Recent structural changes have intensified the relevance of these issues. Rising educational attainment, increased migration, and population ageing have transformed European labour

markets over the past decades. At the same time, the COVID 19 pandemic introduced unprecedented disruptions, disproportionately affecting women, younger workers, and individuals in nonstandard employment. These shocks highlighted existing inequalities and raised new questions about labour market resilience and institutional capacity.

In this context, comparative analysis using harmonized cross national data is essential. By examining employment determinants across multiple European countries during the period 2020–2022, this study aims to identify which individual level factors consistently shape employment outcomes and which are mediated by institutional arrangements. The next section builds on this motivation by clearly defining the research problem and outlining the specific gaps in the existing literature that this thesis seeks to address.

1.2 Research Problem and Significance

This section defines the central research problem addressed by this thesis and explains why it is of both academic and policy relevance. It argues that existing research has not sufficiently explained how individual employment determinants interact with institutional contexts across European countries, particularly during periods of economic disruption.

A substantial literature examines the relationship between individual characteristics and employment outcomes. Education, work experience, gender, health status, and migration background are consistently identified as key predictors of labour market participation (Becker, 1964; Blau & Kahn, 2013; Chiswick & Miller, 2009). Parallel research highlights the role of institutions, including welfare regimes, labour market regulations, and family policies, in shaping aggregate employment levels and inequalities (Esping Andersen, 1990; OECD, 2018). However, these two strands of research are often treated separately. As a result, limited attention is paid to how individual level characteristics translate into employment outcomes differently across institutional settings.

The core research problem addressed in this thesis is the lack of integrated, comparative evidence on employment determinants across Europe. Many existing studies focus on single countries or small groups of countries, which restricts their ability to distinguish between universal labour market mechanisms and context specific effects. When cross country analyses are conducted, they often rely on highly aggregated indicators or narrow subsets of the population. This limits insight into how gender, family structure, immigration status, and health interact with national institutions to shape employment outcomes.

Methodological limitations further compound this problem. Much of the empirical literature relies on standard regression frameworks without systematically testing the robustness of results across model specifications or population subgroups. Non linearities, interaction effects, and heterogeneity are frequently underexplored, despite their theoretical relevance.

As a consequence, empirical findings are sometimes difficult to generalize or translate into policy relevant conclusions.

The period covered by the European Social Survey Round 10 intensifies the importance of addressing these gaps. The years 2020–2022 were marked by unprecedented labour market disruptions linked to the COVID 19 pandemic. Employment shocks were unevenly distributed across sectors and demographic groups, and institutional responses varied substantially across countries. Studying employment determinants during this period offers a unique opportunity to assess how resilient different institutional arrangements are in mediating individual labour market risks.

This thesis addresses these limitations by providing a comprehensive, micro level analysis of employment outcomes across 31 European countries using harmonized survey data. By explicitly linking individual characteristics to institutional contexts, the study aims to identify which employment determinants are stable across countries and which are contingent on policy environments. The following section translates this research problem into clear research objectives that guide the empirical analysis.

1.3 Research Objectives

This section outlines the specific objectives of the thesis. The objectives translate the research problem into a structured analytical agenda and define the scope of the empirical investigation. Each objective is designed to be empirically testable and directly linked to the theoretical framework and data employed in the study.

The first objective is to identify the individual level determinants of employment across European countries. The analysis focuses on core characteristics highlighted in labour economics, including educational attainment, age, gender, family structure, health status, and migration background. By examining these factors simultaneously, the study aims to assess their relative importance in shaping employment outcomes and to evaluate whether commonly observed relationships hold across a diverse set of national labour markets.

The second objective is to examine heterogeneity in employment determinants across population subgroups. Employment risks are not evenly distributed, and prior research suggests that the effects of education, family responsibilities, and health constraints differ substantially by gender, age, and migration status (Blau & Kahn, 2013; Kogan, 2011). This thesis seeks to quantify these differences by conducting subgroup analyses that reveal how employment probabilities vary within and between key demographic groups.

The third objective is to analyse cross country variation in the relationship between individual characteristics and employment outcomes. Rather than treating country

differences as residual variation, the study explicitly investigates how national institutional contexts condition individual level effects. This objective addresses whether similar individuals face systematically different employment prospects depending on welfare regime characteristics, labour market regulations, and family policy environments.

The fourth objective is to assess the extent to which institutional arrangements moderate employment inequalities. The analysis examines whether countries with more extensive social protection, childcare provision, and inclusive labour market policies exhibit smaller employment gaps related to gender, family status, health limitations, and migration background. This objective directly links micro level outcomes to macro level policy environments.

The fifth objective is methodological. The thesis aims to strengthen empirical inference by combining traditional econometric models with complementary machine learning techniques. While econometric models are used for interpretation and hypothesis testing, machine learning approaches serve to assess model stability, capture non linear relationships, and validate key findings. This dual strategy is intended to enhance robustness rather than replace theory driven analysis.

Together, these objectives position the thesis to make a substantive contribution to the literature on employment and labour market inequality in Europe. The next section derives explicit research questions from these objectives, providing a clear link between theory, data, and empirical strategy.

1.4 Research Questions

This section formulates the research questions that guide the empirical analysis. The questions are derived directly from the research objectives and are designed to capture both general employment patterns and context specific differences across European labour markets. Together, they structure the investigation of how individual characteristics and institutional environments interact to shape employment outcomes.

The first research question addresses the core relationship between individual characteristics and employment. Specifically, it asks which personal attributes are most strongly associated with employment probability across European countries during the period 2020–2022. The analysis focuses on education, age, gender, family structure, health status, and migration background, all of which are measured consistently within the European Social Survey. This question aims to establish a baseline understanding of employment determinants using harmonized micro level data.

The second research question examines whether the effects of these individual characteristics are uniform across population groups. Employment outcomes often differ substantially between men and women, between parents and non parents, and between native born individuals and immigrants. This question therefore asks how the relationship between individual characteristics and employment varies across key demographic subgroups. Particular attention is paid to gender differences in the presence of children, as well as to employment gaps associated with migration background and self reported health limitations.

The third research question focuses explicitly on cross country variation. It asks whether individuals with similar characteristics face different employment prospects depending on the country in which they reside. This question moves beyond average effects to investigate how national contexts shape individual labour market outcomes. By incorporating country level information and comparative analysis, the study assesses whether observed employment differences reflect institutional arrangements rather than individual deficits.

The fourth research question examines the moderating role of institutions in shaping employment inequalities. It asks whether welfare regimes, labour market regulations, and family policy environments influence the strength of individual level employment determinants. For example, the analysis investigates whether gender and family related employment gaps are smaller in countries with extensive childcare provision, or whether health related employment penalties are mitigated in contexts with stronger social protection systems.

The fifth research question addresses the robustness and stability of empirical findings. It asks whether conclusions drawn from standard econometric models remain consistent when alternative specifications and complementary machine learning techniques are applied. This question is particularly relevant given the complex, potentially non linear relationships between individual characteristics and employment outcomes, as well as the heterogeneous nature of European labour markets.

Taken together, these research questions provide a structured framework for analysing employment determinants across Europe. They ensure a clear link between theory, data, and methodology, and they prepare the ground for the formulation of explicit research hypotheses in the following section.

1.5 Research Hypotheses

This section presents the research hypotheses tested in the thesis. The hypotheses are derived directly from the theoretical framework and are formulated to be empirically testable using European Social Survey data. Each hypothesis specifies the expected relationship

between key variables and employment outcomes, while acknowledging the role of institutional contexts in shaping these relationships.

H1: Human Capital and Employment

The first hypothesis concerns the relationship between human capital and employment. Human capital theory predicts that higher levels of education increase individual productivity and, consequently, the likelihood of employment (Becker, 1964; Mincer, 1974). Empirical studies across European countries consistently show positive associations between educational attainment and employment probabilities, although the magnitude of this relationship varies across contexts.

H1: Individuals with higher levels of educational attainment are more likely to be employed than individuals with lower levels of education across European countries.

This hypothesis is tested using educational attainment as measured by the International Standard Classification of Education (ISCED). The expectation is a monotonic relationship, with tertiary education associated with the highest employment probabilities.

H2: Gender Differences in Employment

The second hypothesis addresses gender differences in employment outcomes. Despite rising female labour force participation, women continue to face lower employment probabilities than men in many European countries. Theoretical explanations point to gendered labour supply decisions, occupational segregation, and discrimination, as well as unequal distribution of unpaid care work (Blau & Kahn, 2013).

H2: Women have lower employment probabilities than men, after controlling for education, age, and other individual characteristics.

This hypothesis reflects average differences across countries and does not assume uniformity. Cross country variation is examined in later analyses.

H3: Family Structure and Employment

The third hypothesis focuses on the role of family responsibilities. Household economics models predict that the presence of children affects labour supply decisions differently for men and women. Empirical research documents motherhood penalties and, in some contexts, fatherhood premiums in employment outcomes.

H3: The presence of dependent children is associated with lower employment probabilities for women, while having no negative or a positive association with employment for men.

This hypothesis is tested through interaction terms between gender and family structure, allowing for differential effects by parental status.

H4: Immigration Background and Employment

The fourth hypothesis examines employment outcomes among individuals with a migration background. Immigration economics highlights the importance of host country specific human capital, language proficiency, and credential recognition. As a result, immigrants often experience lower employment probabilities than native born individuals, particularly shortly after arrival.

H4: Individuals with an immigrant background have lower employment probabilities than native born individuals, holding constant education and demographic characteristics.

Where data allow, heterogeneity by duration of residence is explored to assess integration patterns.

H5: Health Constraints and Employment

The fifth hypothesis addresses the role of health in shaping employment outcomes. Health limitations reduce work capacity and may lead to discrimination or exclusion from the labour market. Prior studies show strong associations between poor health and non employment across Europe.

H5: Individuals reporting health limitations or disabilities have lower employment probabilities than those without such limitations.

This hypothesis accounts for both direct productivity effects and institutional mediation through disability and social protection systems.

H6: Institutional Moderation of Employment Determinants

The final hypothesis integrates the preceding hypotheses by introducing institutional context. Institutional theory predicts that welfare regimes, labour market policies, and family support systems shape how individual characteristics translate into employment outcomes.

H6: The strength of individual level employment determinants varies systematically across countries, with smaller employment gaps related to gender, family structure, health, and migration background in countries with more inclusive institutional arrangements.

This hypothesis is tested through cross country comparisons and interaction effects between individual characteristics and institutional indicators.

Taken together, these hypotheses form a coherent analytical framework linking individual characteristics to employment outcomes within diverse institutional contexts. The next

section specifies the scope and limitations of the study, clarifying the assumptions under which these hypotheses are tested.

1.6 Scope and Limitations

This section clarifies the scope of the study and discusses its limitations. Defining the boundaries of the analysis is essential for interpreting the results correctly and for distinguishing between what the study can explain and what lies beyond its empirical reach.

The empirical scope of the thesis is limited to European countries participating in the European Social Survey Round 10, covering the period from 2020 to 2022. The analysis includes 31 countries and focuses on individuals of working age. The use of a harmonized, cross national survey allows for direct comparison of employment determinants across diverse institutional contexts, which is a key strength of the study. However, the focus on Europe means that findings cannot be directly generalized to non European labour markets with substantially different institutional arrangements, such as those in low or middle income countries.

The study concentrates on employment status as the primary outcome variable. Employment is measured using self reported current labour market status, which is standard in comparative labour market research (OECD, 2018). While this measure captures whether individuals are employed, it does not provide detailed information on job quality, working hours, contract type, or earnings. As a result, the analysis focuses on access to employment rather than on employment conditions or wage inequality, which are important but distinct dimensions of labour market outcomes.

A further limitation relates to the cross sectional nature of the data. The European Social Survey does not follow individuals over time, which restricts the ability to make strong causal claims about the effects of education, family transitions, or health changes on employment. Reverse causality remains a potential concern, particularly in the case of health and employment, where non employment may also contribute to deteriorating health (Currie & Madrian, 1999). The analysis addresses this issue by careful model specification and robustness checks, but causal interpretations are necessarily cautious.

Measurement limitations also warrant attention. Key variables such as health status and disability are based on self reports, which may be influenced by cultural norms and reporting biases across countries. Prior research suggests that cross national differences in self rated health partly reflect differences in expectations and social context (Jürges, 2007). Similarly, migration background is measured using country of birth, which does not capture legal status, language proficiency, or recognition of foreign qualifications in detail. These

limitations are common in comparative survey research and are acknowledged in the interpretation of results.

Finally, the study does not explicitly model macroeconomic shocks or short term policy changes beyond what is implicitly captured by country level effects. While the period of analysis coincides with the COVID 19 pandemic, the ESS data do not allow for precise identification of pandemic specific mechanisms. Instead, the study focuses on how existing institutional frameworks mediated employment outcomes during a period of heightened labour market stress.

Despite these limitations, the scope of the analysis is appropriate for addressing the research questions posed. By combining detailed individual level data with a comparative institutional perspective, the study provides robust insights into employment determinants across Europe. The following section outlines the structure of the thesis and explains how each chapter contributes to the overall argument.

1.7 Structure of the Thesis

This section outlines the structure of the thesis and explains how each chapter contributes to the overall argument. The thesis is organized to move from conceptual foundations to empirical analysis and, finally, to interpretation and policy implications. Each chapter builds on the previous one, ensuring coherence between theory, data, and results.

Chapter 1 introduces the research problem and motivates the study. It establishes the importance of examining employment determinants across Europe through a comparative lens and clearly defines the research objectives, questions, and hypotheses. By situating the analysis within recent structural changes in European labour markets, including the COVID 19 period, this chapter provides the conceptual and empirical rationale for the study.

Chapter 2 develops the theoretical framework that underpins the empirical analysis. It draws on established theories in labour economics and social policy, including human capital theory, labour supply models, discrimination theories, immigration economics, health and disability frameworks, and institutional theory. Each theoretical perspective is discussed in terms of its core mechanisms, empirical predictions, and relevance for employment outcomes. The chapter culminates in an integrated analytical model that links individual characteristics to employment outcomes within institutional contexts and explicitly connects theoretical expectations to the hypotheses formulated in Chapter 1.

Chapter 3 reviews the empirical literature on employment determinants in Europe. It synthesizes evidence from peer reviewed journal articles, working papers, and policy oriented research published in leading outlets indexed in databases such as Google Scholar,

RePEc/IDEAS, and major journal platforms. The review critically evaluates methodological approaches, data sources, and key findings related to education, gender, family structure, immigration, health, and institutional variation. Rather than providing a descriptive overview, the chapter identifies unresolved debates and empirical gaps that the thesis seeks to address.

Chapter 4 describes the data and methodology. It introduces the European Social Survey Round 10, outlines the sampling design and country coverage, and defines all variables used in the analysis. The chapter explains the econometric strategy, including model specification, estimation techniques, and robustness checks. It also justifies the complementary use of machine learning methods as validation tools, clarifying their role in assessing non linearities and model stability rather than causal inference.

Chapter 5 presents the empirical results. It reports baseline regression findings, hypothesis tests, and marginal effects, followed by subgroup analyses and cross country comparisons. Results are presented in clearly structured tables and figures that can be interpreted independently of the text. Robustness checks and alternative specifications are discussed to assess the sensitivity of the findings.

Chapter 6 discusses the results in relation to the theoretical framework and existing literature. It interprets the empirical findings, highlights areas of convergence and divergence with prior research, and explores their implications for understanding employment inequalities in Europe. Policy implications are discussed in light of institutional differences, and limitations are revisited.

Chapter 7 concludes the thesis by summarizing the main contributions and outlining directions for future research. It reflects on the theoretical, empirical, and policy relevance of the findings and situates the study within broader debates on labour market inequality and social policy.

From a broader perspective, the thesis aims to contribute to labour economics by providing a systematic, comparative analysis of employment determinants across Europe using harmonized micro level data. The remaining work in Chapters 1–3 involves finalizing the theoretical synthesis, deepening the empirical literature review with precise references, and ensuring that all hypotheses are tightly linked to both theory and empirical strategy. Subsequent chapters will operationalize these foundations through rigorous data analysis and interpretation.

Chapter 2: Theoretical Framework

2.1 Introduction to Theoretical Foundations

This chapter establishes the theoretical basis for understanding employment determinants across European countries. It argues that individual labour market outcomes emerge from the interaction of personal characteristics with institutional structures. By integrating multiple economic and social theories, the chapter provides a conceptual model that explains why education, gender, family structure, migration background, and health influence employment differently across countries. The theoretical framework also informs the hypotheses, guides the empirical strategy, and provides a benchmark for interpreting results.

The first part of the chapter focuses on human capital theory, which posits that investments in education and skills increase productivity and, consequently, labour market opportunities (Becker, 1964; Mincer, 1974). Education enhances both knowledge and cognitive skills, making individuals more attractive to employers. However, the returns to human capital are context dependent. For example, immigrants may possess educational credentials that are not fully recognized in the host country, reducing the expected employment benefit (Chiswick & Miller, 2009). Similarly, the age at which education is obtained, the field of study, and the alignment with labour market demand can significantly alter employment outcomes. Empirical studies show that tertiary education is associated with higher employment probabilities across Europe, but the magnitude of this effect varies by country and sector (OECD, 2018).

The second part examines labour supply theory, which conceptualizes employment as the outcome of individual decisions balancing labour and leisure, subject to constraints. The labour-leisure model suggests that participation depends not only on potential earnings but also on household responsibilities, preferences, and opportunity costs (Killingsworth, 1983). For example, the presence of young children disproportionately affects women's labour supply in countries with limited childcare provision, while men are generally less affected. Labour supply theory also informs the analysis of age related employment patterns, as older workers may face diminishing marginal returns to labour or health related constraints.

A third theoretical lens is discrimination theory, which addresses systematic barriers that affect employment outcomes beyond individual productivity. Taste based discrimination (Becker, 1957) and statistical discrimination (Arrow, 1973; Phelps, 1972) can result in persistent employment gaps by gender, ethnicity, or migration status. For instance, employers may favor candidates from majority groups or penalize applicants perceived as higher risk due to family obligations or foreign credentials. Integrating discrimination theory

allows the study to examine why two equally qualified individuals may face different employment prospects in the same country.

Finally, institutional theory emphasizes the role of national policies and welfare regimes in shaping labour market outcomes (Esping Andersen, 1990). Employment is influenced not only by individual characteristics but also by childcare systems, parental leave policies, social protection schemes, and labour regulations. Institutional arrangements can amplify or mitigate the effects of human capital, gender, and health limitations, providing a framework for understanding cross country heterogeneity in employment outcomes.

In sum, this chapter establishes that employment outcomes are multidimensional and context dependent. By combining human capital, labour supply, discrimination, and institutional theories, the thesis develops an integrated framework for testing the hypotheses outlined in Chapter 1. The next section presents human capital theory in detail, including its empirical predictions and relevance for the European context.

2.2 Human Capital Theory

2.2.1 Foundations (Becker, Mincer, Schultz)

Human capital theory posits that investments in education and training enhance individual productivity, which subsequently increases employment prospects and earnings. Gary Becker (1964) formalized the idea that skills and knowledge acquired through formal schooling, vocational training, or on the job learning constitute an economic asset. Becker emphasized that education is not merely a personal good but a productive resource, generating returns to both individuals and society. The framework distinguishes between general human capital, which is transferable across employers, and specific human capital, which is firm or context specific, shaping employment mobility and wage growth.

Jacob Mincer (1974) extended this framework by explicitly modelling the relationship between schooling, labour market experience, and earnings. Mincer demonstrated that both the quantity and quality of education matter and introduced the well known Mincerian earnings function, where years of schooling and cumulative experience predict labour market outcomes. This model remains a cornerstone for empirical studies on returns to education and underpins many econometric approaches used to analyse employment probabilities.

Schultz (1961) emphasized a broader societal perspective, arguing that investments in human capital yield collective productivity gains. According to Schultz, educated workers contribute not only through individual productivity but also via spillover effects, such as innovation diffusion and skill accumulation within the economy. This theoretical foundation

highlights why education can shape labour market outcomes beyond the individual level, particularly in countries with strong knowledge intensive sectors.

Together, these foundational theories provide a robust rationale for focusing on education as a primary determinant of employability. They also emphasize that returns to education are contingent on both individual attributes and external conditions, including institutional frameworks and labour market structure. By integrating these insights, the thesis establishes a strong theoretical link to Hypothesis H1, which predicts that higher educational attainment increases employment probability, and H6, which anticipates variation across countries due to institutional factors.

2.2.2 Education and Employment Nexus

Education shapes employability through multiple interrelated mechanisms. At its core, schooling enhances cognitive and technical skills, equips individuals with problem solving abilities, and provides knowledge directly applicable to the labour market (Hanushek & Woessmann, 2008). Cognitive skills, such as numeracy, literacy, and analytical reasoning, improve task performance across sectors and increase adaptability in rapidly changing economic environments. Technical and field specific skills further align individual capabilities with employer demand, enabling higher productivity and more consistent employment outcomes.

Education also functions as a signal to employers. According to Spence's signalling theory (1973), educational attainment communicates underlying traits discipline, persistence, and cognitive capacity that may not be directly observable. Even when formal skills are imperfectly matched to the job, higher education indicates potential productivity, reducing information asymmetry between employer and employee. This signalling effect helps explain why individuals with comparable work experience but different educational backgrounds may experience divergent employment outcomes.

The interaction between education and labour market demand further conditions employability. In countries with knowledge intensive economies, such as Sweden, Germany, and the Netherlands, tertiary education aligns closely with sectoral needs, resulting in higher employment probabilities for graduates (OECD, 2018). In contrast, in countries with rigid labour markets or slower structural transformation, such as Italy or Spain, graduates may face underemployment or longer periods of unemployment, demonstrating that educational returns are context dependent (Cappellari & Jenkins, 2004; Eurostat, 2021). These findings underscore the need to integrate institutional variables into analyses of educational impacts, as highlighted in Hypothesis H6.

Empirical research consistently supports the positive link between education and employment. Cross European studies indicate that individuals with tertiary education are 15–25% more likely to be employed than those with only secondary education, after controlling for age, gender, and health status (Blau & Kahn, 2013; OECD, 2018). Moreover, the magnitude of this effect varies systematically with country specific policies. For example, recognition of foreign qualifications enhances employment prospects for immigrants, while labour market segmentation and rigid hiring practices reduce returns to education (Dustmann et al., 2016; Kogan, 2011). These findings directly inform Hypothesis H1, predicting that higher educational attainment increases employment probability, and Hypothesis H6, which anticipates cross country variation in these effects due to institutional contexts.

Education also interacts with other individual characteristics, producing heterogeneous effects. Women in countries with extensive childcare provision benefit more from higher education, as they can maintain continuous labour force attachment, reducing career interruptions (Gornick & Meyers, 2003; Thévenon, 2011). Immigrants with host country specific education or recognized foreign credentials achieve substantially better integration outcomes, illustrating the interplay of human capital, institutional support, and discrimination mechanisms.

In conclusion, the education employment nexus demonstrates that schooling enhances employability through skill accumulation, signalling, and alignment with labour market demand. The effect is moderated by country specific institutions, labour market structures, and policy frameworks. By integrating these mechanisms, this section provides a clear theoretical foundation for testing H1 and H6 in a comparative European context.

2.2.3 Empirical Predictions

Building on the theoretical foundations and the education employment nexus, this section translates human capital theory into empirically testable predictions for European labour markets. It specifies the expected magnitude and direction of educational effects on employment, identifies moderating factors at the country level, and provides evidence from previous studies to justify these expectations.

Prediction 1: Positive effect of educational attainment on employment probability (H1). Human capital theory implies that higher levels of education increase employability by enhancing skills, productivity, and signalling competence to employers (Becker, 1964; Mincer, 1974; Spence, 1973). Empirical evidence across Europe supports this mechanism: individuals with tertiary education are 15–25% more likely to be employed than those with secondary education (Blau & Kahn, 2013; OECD, 2018). Similarly, micro level analyses show that each additional year of schooling raises employment probability by 2–3 percentage

points, controlling for age, gender, and health (Cappellari & Jenkins, 2004). Therefore, we predict that, all else equal, higher educational attainment will significantly increase the likelihood of being employed in the sample.

Prediction 2: Cross country heterogeneity of education returns (H6). Institutional contexts shape how education translates into employment. Welfare regimes, labour market flexibility, and recognition of foreign qualifications can amplify or dampen the benefits of human capital (Esping Andersen, 1990; Dustmann et al., 2016; Kogan, 2011). For example, in social democratic countries like Sweden and Denmark, the alignment of education with labour demand and comprehensive social policies allows graduates to achieve near universal employment, while in Southern European countries such as Italy or Spain, rigid labour markets and weak credential recognition reduce returns (Eurostat, 2021; OECD, 2018). Accordingly, we predict that the positive effect of education on employment will be stronger in countries with flexible labour markets and inclusive institutional frameworks, and weaker in restrictive contexts.

Prediction 3: Interaction effects with gender and migration status. Human capital returns are not uniform across all groups. Women may experience lower returns to education in countries with limited childcare provision, while immigrants may face reduced benefits if foreign credentials are undervalued (Gornick & Meyers, 2003; Dustmann et al., 2016). Therefore, we expect that the employment advantage of higher education will be moderated by gender and migration status, interacting with institutional variables to produce heterogeneous outcomes across countries.

Prediction 4: Policy and institutional mediation. Institutional support such as recognition of foreign qualifications, active labour market programs, and anti discrimination policies mediates the translation of human capital into employment (OECD, 2018; Mandel & Semyonov, 2006). Countries with strong integration policies and flexible labour markets enable higher returns to education for both women and immigrants, while restrictive policies limit these returns. Consequently, we predict that institutional variables will significantly moderate the effect of education on employment, providing empirical justification for including cross level interactions in the model.

In sum, these empirical predictions operationalize human capital theory within the European context. They provide measurable expectations for the effect of education on employment (H1), while highlighting the conditional role of institutions (H6) and interactions with gender and migration. The next section, 2.3 Labour Supply Theory, builds on these insights by incorporating household and time allocation constraints, further explaining variations in employment outcomes beyond education.

2.3 Labour Supply Theory

2.3.1 Labour–Leisure Choice Model

Labour supply theory provides a microeconomic framework to understand employment as the outcome of individual decision making, balancing work and leisure under constraints of income and time. The labour–leisure choice model formalizes the trade off between labour market participation and non working activities, emphasizing that employment decisions are not determined solely by human capital but also by preferences, household responsibilities, and institutional context (Killingsworth, 1983; Blundell & MaCurdy, 1999).

In this model, individuals allocate a finite amount of time between labour and leisure to maximize utility. Utility depends both on the consumption enabled by labour income and the intrinsic value of leisure time, which includes rest, caregiving, education, and personal activities. The optimal labour supply occurs when the marginal utility of leisure equals the marginal utility of income derived from additional work hours (Hamermesh, 1993). This approach explains why two individuals with similar education or skills may exhibit different employment patterns: household constraints, personal preferences, and policy environments shape the choice between working and non working time.

Empirical evidence demonstrates that the labour–leisure trade off is particularly relevant for women and primary caregivers. For instance, Blau and Kahn (2013) show that women’s labour participation is significantly affected by the presence of young children and availability of childcare services. In countries with limited childcare provision, many highly educated women reduce their labour supply or leave the labour market temporarily, even though their human capital would predict high employability. Conversely, in Scandinavian countries with extensive childcare and flexible work arrangements, women’s labour supply is less constrained, and education translates more consistently into employment.

The labour–leisure framework also highlights the interaction with institutional policies. Parental leave, subsidized childcare, flexible working hours, and part time employment options reduce the opportunity cost of labour participation, increasing the likelihood of employment, particularly for women and older workers (Gornick & Meyers, 2003; Thévenon, 2011). This institutional mediation aligns directly with Hypothesis H6, which predicts cross country variation in how individual characteristics translate into employment outcomes. At the same time, Hypothesis H3 draws on this model to predict that family responsibilities, such as caring for children or elderly relatives, negatively affect employment probability unless mitigated by supportive policies.

In conclusion, the labour–leisure choice model provides a foundational explanation for heterogeneity in employment patterns beyond human capital. By incorporating preferences,

household responsibilities, and institutional constraints, it captures why employment outcomes vary not only between individuals with similar skills but also across countries. This framework establishes the basis for the next sub subsection, 2.3.2 Reservation Wage Theory, which further formalizes the thresholds at which individuals are willing to supply labour under varying economic and policy conditions.

2.3.2 Reservation Wage Theory

The reservation wage is a central concept in labour supply theory, representing the minimum wage at which an individual is willing to accept employment (Burtless, 1986; Killingsworth, 1983). It captures the threshold where the benefits of working, in terms of income and career development, outweigh the utility of non working activities, including leisure, caregiving, and personal time. Understanding reservation wages is critical for explaining why individuals with comparable skills or education may make different labour market choices, particularly in the presence of household constraints or institutional differences.

Gender and family structure effects play a crucial role in shaping reservation wages. Women, especially those with young children, often face higher effective reservation wages because non labour commitments childcare, household management, and caregiving reduce the net benefit of employment (Budig & England, 2001; Blau & Kahn, 2013). For example, in Southern Europe, where public childcare provision is limited, many highly skilled women postpone labour market entry or reduce working hours to balance family responsibilities, effectively raising their reservation wage. Men's labour supply, by contrast, is less constrained by family obligations; in some contexts, fatherhood may even lower their reservation wage due to social norms or income incentives. This differential contributes to observed gender gaps in employment, providing theoretical support for Hypothesis H3.

Migration specific considerations further influence reservation wages. Immigrants often face barriers such as non recognition of foreign qualifications, language limitations, or discrimination, which reduce the expected returns to work (Chiswick & Miller, 2009; Dustmann et al., 2016). Consequently, immigrants may set higher reservation wages, delaying labour market entry or accepting only specific types of employment that compensate for perceived risks or skill underutilization. The interaction between migration status and reservation wages highlights the necessity of integrating both individual and institutional factors in empirical analysis.

Cross country institutional moderation plays a pivotal role in shaping reservation wages. Welfare regimes, social transfers, unemployment benefits, and family support policies alter the opportunity cost of work, thereby influencing labour supply decisions (Esping Andersen, 1990; Gornick & Meyers, 2003). For instance, in Scandinavian countries, generous childcare, paid parental leave, and flexible work policies reduce the effective reservation wage for

women, increasing female labour participation and diminishing employment gaps. Conversely, in liberal or conservative corporatist regimes with weaker support systems, higher reservation wages for women and immigrants result in lower employment probabilities, consistent with Hypothesis H6.

Empirical studies confirm these theoretical predictions. Cross national analyses show that family responsibilities and welfare support jointly determine women's labour force participation, with the highest employment among educated women in countries offering extensive childcare and parental leave (OECD, 2018; Mandel & Semyonov, 2006). For immigrants, recognition of foreign credentials and access to integration programs significantly reduce reservation wages, accelerating labour market integration (Kogan, 2011; Dustmann et al., 2016).

In conclusion, reservation wage theory provides a precise microeconomic mechanism explaining individual variation in labour supply. By integrating household responsibilities, gender norms, migration specific barriers, and institutional support, it clarifies why employment outcomes vary across individuals and countries. This framework complements the labour-leisure choice model and establishes the foundation for 2.3.3 Application to Employment Decisions, linking theory to measurable labour market behaviour.

2.3.3 Application to Employment Decisions

Labour supply theory, incorporating the labour-leisure trade off and reservation wage concepts, provides a structured explanation for observed employment patterns across Europe. Individual decisions to participate in the labour market reflect a combination of personal preferences, household responsibilities, and economic incentives, all shaped by institutional contexts. Understanding these mechanisms allows the prediction of which groups are more likely to work and how policies alter these outcomes.

Women and family responsibilities. Empirical studies consistently show that women's labour supply is strongly influenced by childcare and domestic obligations. In countries with limited institutional support, such as Italy and Spain, women with young children are substantially less likely to participate in the labour market, even when highly educated, reflecting higher effective reservation wages (Budig & England, 2001; Mandel & Semyonov, 2006). Conversely, Scandinavian countries with comprehensive childcare systems and parental leave programs enable women to maintain continuous employment, illustrating how labour supply constraints are mediated by policy. These patterns directly support Hypothesis H3, which predicts that family structure affects employment probability, and highlight that institutional variables (H6) moderate this effect.

Immigrants. Labour supply decisions among immigrant populations often differ from native born individuals due to additional barriers such as recognition of foreign qualifications, language proficiency, and discrimination (Dustmann et al., 2016; Kogan, 2011). These factors raise the effective reservation wage for immigrants, leading some to postpone employment or accept only jobs aligned with prior skills and credentials. Integration policies, language training, and supportive labour market programs reduce these constraints, lowering reservation wages and increasing labour participation. This demonstrates the joint importance of individual characteristics and institutions in shaping employment outcomes, consistent with H3 and H6.

Quantitative expectations. Labour supply models predict measurable effects of gender, family responsibilities, and migration status on employment probabilities. Cross European surveys, including the European Social Survey, show that women with young children have employment probabilities 10–20% lower than childless women in countries with minimal childcare support (OECD, 2018). Similarly, first generation immigrants exhibit employment gaps ranging from 5% in inclusive welfare states to over 15% in restrictive regimes (Dustmann et al., 2016). These quantitative estimates highlight the predictive power of labour supply models when combined with cross country institutional indicators.

Cross country institutional effects. Labour supply theory underscores that policies directly influence the trade off between work and non working activities. Flexible working arrangements, subsidized childcare, and generous parental leave reduce opportunity costs, increasing labour participation among women and secondary earners (Gornick & Meyers, 2003; Thévenon, 2011). In contrast, rigid labour markets and limited social support elevate reservation wages, lowering employment rates. Comparative studies reveal that institutional frameworks explain substantial variation in labour force participation across countries, particularly among vulnerable groups such as mothers and immigrants (Esping Andersen, 1990; Mandel & Semyonov, 2006).

In conclusion, the application of labour supply theory to employment decisions integrates individual preferences, household responsibilities, and institutional contexts. By providing quantitative predictions for women, immigrants, and families, this framework not only explains observed employment patterns but also links directly to Hypothesis H3, which emphasizes the effect of family structure on labour participation, and Hypothesis H6, which anticipates cross country variation due to institutional moderation. These insights form a critical foundation for the subsequent discussion on Discrimination Theories, where additional barriers to employment are explored.

2.4 Discrimination Theories

2.4.1 Taste Based Discrimination (Becker)

Taste based discrimination, as formulated by Gary Becker (1957), proposes that employers, coworkers, or customers may hold preferences or “tastes” against certain groups, such as women, immigrants, or ethnic minorities. Unlike statistical discrimination, which relies on perceived productivity differences, taste based discrimination imposes non economic biases: employers may consciously or unconsciously avoid hiring individuals from certain groups, even if their productivity is comparable to other candidates. This framework provides a microeconomic explanation for persistent employment gaps that cannot be accounted for by education, skills, or experience alone.

Becker’s model posits that discriminatory employers incur a utility cost when hiring preferred groups but may be willing to accept lower profits to avoid working with or hiring certain individuals. Consequently, equally qualified candidates from stigmatized groups are offered lower wages, fewer promotions, or reduced employment opportunities. Empirical studies have confirmed these effects. Bertrand and Mullainathan (2004), using a large scale field experiment in the United States, found that resumes with “ethnic sounding” names received 50% fewer callbacks than identical resumes with “native sounding” names, demonstrating clear taste based bias. While this study is U.S. focused, similar mechanisms operate in European labour markets: OECD (2018) reports that immigrants from non Western countries in Europe are consistently disadvantaged in job acquisition, even after controlling for human capital.

Taste based discrimination is particularly relevant to gender and migration. Women in traditionally male dominated sectors may experience reduced hiring probabilities or wage penalties due to employer preferences, regardless of qualifications (Blau & Kahn, 2013). Similarly, first generation immigrants often encounter barriers based on cultural or linguistic stereotypes rather than observable productivity. These biases interact with institutional frameworks: countries with strong anti discrimination laws, affirmative hiring policies, or inclusive labour market practices exhibit smaller employment gaps, highlighting how policy can mitigate the impact of taste based biases (OECD, 2018; Dustmann et al., 2016).

Linking to your research hypotheses, taste based discrimination underpins **Hypothesis H2**, which predicts persistent gender gaps in employment, and **H4**, which anticipates employment disadvantages for immigrants. Furthermore, the variation in discriminatory effects across countries supports **H6**, which suggests that institutional arrangements moderate the influence of individual level characteristics on labour market outcomes.

In conclusion, taste based discrimination illustrates how non productivity related preferences can generate persistent inequalities in employment. By integrating this theory into the broader framework, the thesis captures mechanisms that operate alongside human capital, labour supply, and institutional factors. Understanding these biases is essential for interpreting cross country differences in employment gaps and provides a foundation for exploring **statistical discrimination**, discussed in the next sub subsection.

2.4.2 Statistical Discrimination (Arrow, Phelps)

Statistical discrimination, as formalized by Arrow (1973) and Phelps (1972), occurs when employers use group level averages to infer individual productivity in situations where personal information is incomplete or costly to obtain. Unlike taste based discrimination, this approach is not driven by personal prejudice but by rational expectations under uncertainty. Employers may assume, for instance, that women with young children are less available for full time work or that immigrants with foreign degrees have skills that are not fully recognized. While individually rational, these practices systematically disadvantage entire groups.

The mechanism operates through employer reliance on observable proxies. For example, if past data indicate that certain demographic groups experience higher turnover, employers may be less willing to hire from these groups, regardless of an individual candidate's true productivity (Arrow, 1973). This leads to a self reinforcing cycle: groups perceived as "risky" experience higher unemployment and underutilization of skills, which in turn confirms employers' initial assumptions (Phelps, 1972; Lang & Lehmann, 2012). In European contexts, studies find that women returning from maternity leave often face delays in reemployment or are offered lower level positions, illustrating statistical assumptions about future availability (Budig & England, 2001; OECD, 2018). Similarly, first generation immigrants frequently encounter reduced employment opportunities due to perceived gaps in human capital or cultural fit (Dustmann et al., 2016; Kogan, 2011).

Statistical discrimination interacts with institutional frameworks, affecting its severity. Countries with robust anti discrimination policies, mandatory credential recognition systems, and active labour market programs can reduce reliance on group level assumptions (OECD, 2018; Esping Andersen, 1990). For example, Germany's formal recognition process for foreign qualifications mitigates employment disadvantages for skilled immigrants, while countries lacking such mechanisms amplify the effects of statistical discrimination. This cross country variation aligns directly with Hypothesis H6, which predicts that institutional arrangements moderate the impact of individual characteristics on employment outcomes.

Gender specific mechanisms are also evident. Employers may infer lower long term commitment or reduced hours from female candidates, especially in societies with traditional gender norms (Blau & Kahn, 2013; Mandel & Semyonov, 2006). These assumptions lead to lower hiring probabilities, reduced wage offers, and slower career progression, supporting Hypothesis H2, which predicts persistent gender gaps. Similarly, immigrants face employment barriers reflecting group level assumptions about skill transferability, linking to Hypothesis H4.

In conclusion, statistical discrimination provides a rational yet inequitable explanation for employment disparities. It complements taste based discrimination by showing that even non prejudiced decision making can produce systematic gaps for women and immigrants. Incorporating this theory into the broader analytical framework enables the thesis to account for individual and group level barriers to employment and to examine how institutional policies can moderate these effects. The next sub subsection, 2.4.3 Gender Specific Mechanisms, will further explore how these theories **manifest in gendered employment patterns across Europe**.

2.4.3 Gender Specific Mechanisms

Gender specific mechanisms provide a lens to understand how structural and social factors intersect with labour market dynamics to create persistent employment inequalities. Despite similar human capital, women often face systematic disadvantages in hiring, promotion, and earnings due to both cultural norms and institutional arrangements (Blau & Kahn, 2013; Budig & England, 2001). These disparities are not solely the result of individual choices but reflect the interaction of gender roles, family responsibilities, and policy environments.

Empirical evidence demonstrates that women experience measurable labour market penalties. For instance, the “motherhood penalty” reduces wages and employment opportunities for women returning from childbirth, even when controlling for education and experience (Budig & England, 2001; Waldfogel, 1997). Conversely, men may benefit from a “fatherhood premium,” reflecting societal expectations of male breadwinning (Glauber, 2008). These gendered patterns highlight the differential returns to human capital across men and women, linking directly to Hypothesis H2, which predicts persistent gender gaps in employment.

Family responsibilities amplify these effects. Women with young children often adjust their labour supply due to childcare obligations, leading to part time employment or temporary withdrawal from the labour market (OECD, 2018; Thévenon, 2011). The magnitude of these constraints varies across countries: Scandinavian nations with subsidized childcare and generous parental leave show smaller employment gaps, whereas Southern European

countries exhibit higher penalties due to limited policy support (Gornick & Meyers, 2003). These cross country variations illustrate the moderating role of institutions, supporting Hypothesis H6.

Gender specific discrimination also interacts with human capital. Even highly educated women face lower employment probabilities in male dominated sectors or when employer assumptions about future family obligations affect hiring decisions (Blau & Kahn, 2013). Statistical discrimination mechanisms exacerbate these patterns: employers may assume women are less committed or available for full time roles, leading to reduced offers or delayed career progression (Arrow, 1973; Budig & England, 2001).

Finally, the cumulative effect of these mechanisms demonstrates that gendered employment gaps are multi layered. They arise from a combination of taste based and statistical discrimination, family related labour supply constraints, and institutional contexts. Countries that combine flexible labour markets with strong social policies tend to reduce these gaps, highlighting the importance of cross country comparisons in understanding labour market inequality (Esping Andersen, 1990; Mandel & Semyonov, 2006).

In conclusion, gender specific mechanisms provide a concrete explanation for persistent differences in employment outcomes between men and women. By integrating evidence from family responsibilities, human capital, and institutional policy, this section strengthens the theoretical foundation for Hypothesis H2 (gender gaps) and H6 (institutional moderation). The next chapter, 2.5 Immigration Economics, will extend this framework to explore employment patterns among immigrants and the interaction of human capital, discrimination, and institutional factors.

2.5 Immigration Economics

2.5.1 Immigrant Labour Market Integration

Immigrant labour market integration is a central concern for understanding employment disparities in Europe. Across countries, first generation immigrants consistently experience lower employment probabilities compared with native born individuals, even when controlling for education, experience, and other human capital factors (Dustmann et al., 2016; Kogan, 2011). These gaps reflect both structural barriers and institutional arrangements, highlighting the complex interplay between individual skills and host country contexts.

Empirical studies indicate that employment gaps vary widely by origin and host country. For example, immigrants from non Western countries in Germany or the Netherlands face higher

unemployment rates and longer job search durations compared with Western European or native populations (OECD, 2018). This heterogeneity arises in part from the transferability of human capital: foreign credentials may not be recognized, and language skills or cultural familiarity often limit access to professional networks (Chiswick & Miller, 2009). Immigrants who acquire host country specific education or participate in language and vocational programs experience significantly higher employment probabilities, demonstrating the critical role of host country human capital in labour market integration (Dustmann & Fabbri, 2003).

Institutional integration policies further moderate immigrant employment outcomes. Countries with structured recognition processes for foreign qualifications, active labour market programs, and targeted anti discrimination measures achieve higher integration rates (Kogan, 2011; OECD, 2018). For instance, Sweden and Denmark provide robust vocational training and subsidized language courses for immigrants, resulting in employment gaps of less than 5% relative to natives. In contrast, Southern European countries with limited institutional support exhibit persistent gaps exceeding 15%, illustrating the cross country variation predicted by Hypothesis H6.

Immigrant labour market integration is not uniform across demographic groups. Gender interacts with migration status, as immigrant women often face compounded disadvantages due to family responsibilities and cultural expectations (Thévenon, 2011; Gornick & Meyers, 2003). This dynamic links directly to Hypothesis H4, which predicts lower employment probabilities for immigrants, and further supports H6, reflecting the moderating role of institutional policies. Moreover, younger immigrants or those entering countries with flexible labour markets show higher integration rates, highlighting the combined influence of individual characteristics, timing, and policy environments.

In conclusion, immigrant labour market integration demonstrates the joint importance of human capital, discrimination, and institutional frameworks in shaping employment outcomes. By accounting for origin, host country specific skills, and supportive policies, this section provides a foundation for understanding variation in immigrant employment and testing Hypotheses H4 and H6. The next sub subsection, 2.5.2 Host Country Specific Human Capital, will further explore how acquiring local skills and credentials influences immigrant employment probabilities across European countries.

2.5.2 Host Country Specific Human Capital

Host country specific human capital refers to skills, knowledge, and qualifications that are recognized and valued within the labour market of the country in which an immigrant resides. Unlike general education or experience acquired abroad, these skills are directly aligned with local labour market norms, institutional rules, and employer expectations

(Chiswick & Miller, 2009). Acquiring host country specific human capital is a critical mechanism for overcoming employment barriers and achieving labour market integration.

The mechanisms of skill transfer and adaptation operate on multiple levels. First, formal recognition of foreign credentials enhances employability by validating prior education, allowing immigrants to compete effectively with native born workers (Kogan, 2011). Second, language proficiency is essential for communication, comprehension of local regulations, and network formation, significantly influencing job search outcomes (Dustmann & Fabbri, 2003). Third, participation in vocational training or local higher education programs provides both technical skills and familiarity with host country work culture, increasing perceived productivity and reducing employer uncertainty. These mechanisms jointly reduce the effective reservation wage for immigrants, increasing their labour market participation, which directly relates to Hypothesis H4.

Quantitative evidence illustrates the impact of host country specific human capital. For instance, in Germany, immigrants who complete vocational training in the host country have employment probabilities 20–25 percentage points higher than those relying solely on foreign qualifications (Kogan, 2011; Dustmann et al., 2016). Similarly, language proficiency in the destination country increases the likelihood of securing skilled employment by 10–15% in several European nations, even after controlling for overall education and work experience (Chiswick & Miller, 2009). These findings confirm that acquiring locally relevant skills significantly enhances employability beyond general human capital.

The interaction with gender and policy further shapes returns to host country specific human capital. Immigrant women often face additional constraints due to family responsibilities or cultural expectations, which can limit participation in training or education programs (Gornick & Meyers, 2003). Countries that provide accessible childcare, subsidized language courses, and targeted vocational programs enable women to convert their skills into employment more effectively, demonstrating cross country institutional moderation consistent with Hypothesis H6. Conversely, restrictive labour markets or lack of integration policies reduce the value of host country specific skills, perpetuating employment gaps.

In conclusion, host country specific human capital is a critical determinant of immigrant labour market integration. By enhancing local skills, credentials, and familiarity with institutional norms, immigrants increase their employment prospects, particularly when supported by inclusive policies. This framework directly informs Hypothesis H4, predicting employment disadvantages for immigrants without recognized local skills, and Hypothesis H6, emphasizing the moderating role of institutional arrangements. The next section, 2.6 Institutional Theory, will extend the analysis to examine how national policies and welfare regimes shape employment outcomes across all groups.

2.6 Institutional Theory

2.6.1 Welfare Regime Typology (Esping Andersen)

Institutional theory emphasizes the role of national policy frameworks, labour market regulations, and social welfare systems in shaping employment outcomes. Esping Andersen's (1990) welfare regime typology remains a cornerstone for understanding cross country variation in labour market integration, gender equality, and immigrant employment. According to this framework, countries are classified into three primary regimes: liberal, conservative corporatist, and social democratic, each with distinct approaches to social protection, labour market flexibility, and family support.

Social democratic regimes, exemplified by Sweden, Denmark, and Norway, combine universal social protection with active labour market policies and extensive family support. These countries provide subsidized childcare, generous parental leave, and flexible working arrangements, enabling high female labour force participation and facilitating employment for immigrants (Gornick & Meyers, 2003; Thévenon, 2011). Empirical evidence demonstrates that in these regimes, employment rates for highly educated women approach those of men, and first generation immigrants integrate more rapidly into the labour market, confirming that institutional support moderates human capital returns (OECD, 2018; Dustmann et al., 2016).

Conservative corporatist regimes, including Germany, France, and Italy, emphasize employment protection and social insurance contributions, often tied to traditional family structures. These policies create strong employment incentives for male workers but limit flexibility for women, particularly mothers, thereby sustaining the motherhood penalty (Mandel & Semyonov, 2006; Budig & England, 2001). Immigrant integration is slower in these countries unless targeted policies exist for credential recognition or vocational training (Kogan, 2011). The interaction between family oriented policies and rigid labour market rules exemplifies how welfare regimes shape employment probabilities differently for subgroups, consistent with H2, H3, and H6.

Liberal regimes, represented by the United Kingdom and Ireland, prioritize market mechanisms with minimal state intervention. Labour markets are flexible, but social protections, childcare subsidies, and active labour market programs are limited. This structure often results in high employment volatility, particularly for women and immigrants, and creates reliance on private resources to balance work and family responsibilities (Esping Andersen, 1990; OECD, 2018). Studies show that in liberal regimes, the gender gap in employment persists despite high educational attainment, reflecting limited institutional mitigation of labour supply constraints (Blau & Kahn, 2013).

Cross country empirical analyses reveal that welfare regime type significantly moderates the effect of individual characteristics on employment outcomes. For example, in Scandinavian countries, tertiary education reliably predicts high employment across genders, while in conservative regimes, the same human capital returns are conditional on household and institutional support (OECD, 2018; Cappellari & Jenkins, 2004). Immigrant employment patterns follow similar trends: integration is faster in social democratic regimes due to proactive policies, whereas restrictive or market driven regimes amplify the disadvantages of foreign credentials (Dustmann et al., 2016).

In conclusion, Esping Andersen's welfare regime typology provides a robust analytical lens to understand institutional moderation of labour market outcomes, directly linking to Hypotheses H2, H3, H4, and H6. It highlights why education, gender, family responsibilities, and immigrant status do not operate in isolation but interact with national policies to shape employment probabilities. The next sub section, 2.6.2 Institutional Effects on Employment, will build on this framework to examine specific policies, including childcare, parental leave, and labour protection, and their measurable impact on employment across Europe.

2.6.2 Institutional Effects on Employment

Institutional arrangements including welfare policies, labour regulations, and family support programs play a critical role in shaping employment outcomes by influencing both the incentives and constraints faced by workers. Beyond individual characteristics such as education, age, or health, institutions determine whether these attributes translate into actual employment opportunities. This section examines policy level mechanisms, including childcare provision, parental leave, and labour market protections, and their measurable impact across European countries.

Childcare provision is among the most influential institutional factors affecting labour supply, particularly for women. Extensive cross national research demonstrates that access to affordable, high quality childcare increases female labour force participation and reduces gendered employment gaps (Gornick & Meyers, 2003; OECD, 2018). For example, in Sweden, subsidized childcare enables women to remain in full time employment following childbirth, minimizing the motherhood penalty. In contrast, Southern European countries such as Italy and Spain, where public childcare coverage is limited, exhibit persistent gaps between men and women, even among highly educated cohorts (Thévenon, 2011; Budig & England, 2001). These patterns directly inform Hypotheses H2 and H3, highlighting the institutional moderation of gender and family related employment outcomes.

Parental leave policies further shape employment trajectories. Generous and flexible leave systems, when combined with job protection, encourage labour market continuity for mothers and fathers (Gornick & Meyers, 2003; OECD, 2018). Empirical evidence shows that

Scandinavian countries with paid parental leave exceeding 12 months maintain high re entry rates for women post childbirth, whereas countries with short or unpaid leave experience significant temporary exits and skill depreciation (Mandel & Semyonov, 2006). Parental leave policies also affect household labour supply decisions and reservation wages, illustrating the interaction between institutional support and individual choices predicted by Hypothesis H6.

Labour market protections, such as employment security, anti discrimination legislation, and minimum wage enforcement, further influence employment outcomes. Conservative corporatist regimes provide strong protections for long term employees, but rigid hiring and firing regulations can reduce labour market flexibility, disproportionately affecting women, immigrants, and youth (Esping Andersen, 1990; Cappellari & Jenkins, 2004). Conversely, liberal regimes prioritize market flexibility but offer fewer safeguards, leading to high employment volatility for vulnerable groups (OECD, 2018). Across regimes, policies that combine protection with active labour market support such as targeted training or integration programs for immigrants enhance overall employment probabilities and reduce disparities (Dustmann et al., 2016; Kogan, 2011).

Cross country comparisons illustrate the moderating effect of institutions. Social democratic regimes consistently display smaller gender and immigrant employment gaps, reflecting coordinated childcare, leave, and labour market policies (Gornick & Meyers, 2003; Thévenon, 2011). Conservative regimes show larger gender penalties but moderate immigrant employment gaps when structured vocational and recognition programs exist (Mandel & Semyonov, 2006; Kogan, 2011). Liberal regimes often exhibit high employment variability and gaps for both women and immigrants, emphasizing the role of policy design in shaping labour market outcomes (OECD, 2018).

In conclusion, institutional policies are critical determinants of employment, mediating the impact of individual characteristics such as gender, family structure, education, and migration status. Childcare, parental leave, and labour protections are measurable mechanisms that shape labour supply, reduce gendered and migration related gaps, and explain cross country variation. These insights directly link to H2, H3, and H6, providing a robust framework to analyze how national policies condition the effects of human capital, family responsibilities, and immigrant status on employment outcomes. The next section, 2.7 Synthesis: Integrated Theoretical Model, will unify human capital, labour supply, discrimination, immigration, and institutional theories into a comprehensive framework for testing all hypotheses (H1–H6) empirically.

2.7 Synthesis: Integrated Theoretical Model

The integrated theoretical model combines insights from human capital, labour supply, discrimination, immigration, and institutional theories to provide a comprehensive framework for understanding employment outcomes across Europe. Each theoretical perspective contributes a distinct mechanism, while their interactions explain the observed heterogeneity in labour market participation, gender gaps, and immigrant integration. By linking these mechanisms explicitly to the research hypotheses (H1–H6), the model offers a structured foundation for empirical testing.

Human capital theory forms the core of the model, positing that educational attainment and skills increase productivity and employment probability (H1) (Becker, 1964; Mincer, 1974; Schultz, 1961). Empirical evidence confirms that higher education consistently predicts higher employment across countries, though returns vary with institutional contexts (H6) (OECD, 2018; Cappellari & Jenkins, 2004). Human capital interacts with labour supply constraints, as individual decisions to participate in the labour market are influenced by family responsibilities, gender roles, and reservation wages (H3) (Budig & England, 2001; Gornick & Meyers, 2003). For example, women with young children or secondary earners face higher effective reservation wages, which may reduce employment probability unless mitigated by supportive policies.

Discrimination theories both taste based (Becker, 1957) and statistical (Arrow, 1973; Phelps, 1972) explain persistent employment gaps not accounted for by human capital. Gender specific mechanisms, such as the motherhood penalty and gendered wage differentials, support H2, while immigrant employment barriers due to perceived productivity or employer bias reinforce H4. Importantly, institutional contexts moderate these effects (H6): countries with robust anti discrimination laws, credential recognition, and inclusive policies reduce the negative impact of bias on women and immigrants (OECD, 2018; Dustmann et al., 2016).

Immigration economics highlights the role of host country specific human capital in integration. Immigrants' education, skills, and experience acquired abroad must be complemented by local qualifications, language proficiency, and vocational training to improve employment outcomes (H4) (Chiswick & Miller, 2009; Kogan, 2011). Labour market policies, including recognition of foreign credentials and active integration programs, further moderate these effects, linking back to H6. Gender interacts with migration status, as immigrant women often experience compounded constraints due to family responsibilities, cultural expectations, and labour market discrimination (H2, H3).

Institutional theory frames the broader context, demonstrating how welfare regimes, labour market protections, childcare, and parental leave shape the translation of individual

characteristics into actual employment outcomes (H6) (Esping Andersen, 1990; Thévenon, 2011; Gornick & Meyers, 2003). Social democratic regimes amplify the returns to human capital for women and immigrants, conservative corporatist regimes moderate gender penalties but maintain structural constraints, and liberal regimes exhibit high variability and reliance on market forces (OECD, 2018; Mandel & Semyonov, 2006). Institutions therefore act as both facilitators and constraints, conditioning the effects of education, family structure, gender, and migration on employment probabilities.

Unified framework. Figure 2.1 illustrates the integrated model: individual level characteristics (education, age, gender, family, health, migration status) interact with labour supply constraints and discrimination mechanisms, while institutional variables (welfare regime, childcare, parental leave, labour protections) moderate these relationships. Each pathway is explicitly linked to the research hypotheses:

- H1: Education increases employment probability (human capital).
- H2: Gender gaps persist, moderated by discrimination and policy.
- H3: Family structure and labour supply constraints influence employment.
- H4: Immigrants face barriers; host country human capital matters.
- H5: Health and disability affect employment (interacts with human capital and policy).
- H6: Institutional arrangements moderate all above relationships across countries.

This integrated model enables a multi level empirical approach, combining individual and country level data to test the relative importance of human capital, discrimination, family structure, immigration, and institutional factors. By unifying theories, it provides a coherent framework for understanding both universal patterns and context specific variations in employment across Europe. Importantly, it bridges the theoretical discussion with the empirical methodology in Chapter 4, where these mechanisms will be operationalized using European Social Survey data.

2.8 Chapter Summary and Link to Empirical Literature

This chapter has established the theoretical foundation for analyzing employment outcomes across Europe, integrating five interrelated frameworks: human capital, labour supply, discrimination, immigration economics, and institutional theory. Each theory contributes distinct insights while interacting to explain observed heterogeneity in employment patterns, gender gaps, immigrant integration, and cross country variation.

Collectively, they provide a coherent lens to operationalize the research hypotheses (H1–H6) in the empirical chapters.

Human capital theory (Becker, 1964; Mincer, 1974; Schultz, 1961) explains the foundational role of education and skills in shaping employability, supporting H1. The education–employment nexus highlights how cognitive, technical, and transferable skills increase labour market participation, while cross country differences underscore the moderating effect of institutions, consistent with H6. Empirical predictions drawn from this framework guide the operationalization of education variables in the dataset.

Labour supply theory (Becker, 1965; Borjas, 2016) provides mechanisms through which household constraints, labour–leisure choices, and reservation wages influence employment. It elucidates why women, parents, and secondary earners may exhibit lower labour force participation even with high human capital, directly informing H3, and how institutional support can mitigate these constraints (H6). Labour supply considerations interact with both discrimination and institutional policies to shape observed outcomes.

Discrimination theories, including taste based (Becker, 1957) and statistical discrimination (Arrow, 1973; Phelps, 1972), explain employment gaps that cannot be attributed to skills alone. Gender specific mechanisms reveal persistent barriers for women (H2) and highlight compounded disadvantages for immigrant women, while statistical assumptions contribute to lower employment probabilities among first generation immigrants (H4). Institutional policies, such as anti discrimination legislation and inclusive labour market programs, moderate these effects, supporting H6.

Immigration economics emphasizes the critical role of host country specific human capital in labour market integration. Immigrants with locally recognized education, language proficiency, and vocational training demonstrate higher employment probabilities, aligning with H4. Cross country institutional variation further conditions these outcomes, consistent with H6, illustrating that integration is a function of both individual skills and supportive policy environments.

Institutional theory, grounded in Esping Andersen’s welfare regime typology (1990), explains macro level differences in employment outcomes. Social democratic, conservative, and liberal regimes create distinct contexts for translating human capital, labour supply decisions, and discrimination into realised employment. Policies such as childcare, parental leave, and labour protections mitigate or exacerbate gendered and migration related gaps, connecting directly to H2, H3, and H6.

Together, these theories form an integrated analytical framework, linking micro level individual characteristics to macro level institutional factors. Figure 2.1 illustrates this multi

level model, which serves as the foundation for the empirical strategy in Chapters 3 and 4. The theoretical review highlights empirical gaps: while previous research confirms individual effects of education, gender, migration, and institutions, few studies simultaneously integrate all mechanisms within a comparative cross country context. This thesis addresses this gap by combining European Social Survey data with country level policy indicators to test H1–H6 rigorously.

In conclusion, Chapter 2 provides a comprehensive conceptual map that bridges theory and empirical analysis. It demonstrates how individual level human capital, household constraints, discrimination, and immigration status interact with institutional arrangements to shape employment. The next chapter, Chapter 3: Review of Empirical Research, will examine existing evidence on these mechanisms, identify patterns and gaps in cross country studies, and justify the empirical approach used to test the hypotheses outlined in this framework.

3. REVIEW OF EMPIRICAL RESEARCH

3.1 Introduction and Review Methodology

This chapter provides a comprehensive review of empirical research on the determinants of employment across Europe, bridging the theoretical framework established in Chapter 2 with the analytical approach of this thesis. The purpose is twofold: first, to summarize and critically assess existing studies on education, gender, family structure, immigration, health, and institutional factors; second, to identify empirical gaps that justify the current research using the European Social Survey (ESS) Round 10 dataset.

The review is guided by the hypotheses (H1–H6) outlined in Chapter 2, ensuring that each section examines mechanisms and evidence relevant to the research questions. The methodology follows a structured approach. First, peer reviewed journal articles, OECD reports, Eurostat statistics, and RePEc working papers were systematically identified using keyword searches (e.g., “employment Europe,” “gender gap labour,” “immigrant integration,” “human capital returns”) across Google Scholar, Scopus, Web of Science, and ResearchGate. Only studies from recognized journals and institutional sources were included to ensure high quality, replicable evidence.

Each subsection follows a consistent analytical structure. The first step is summary of findings, highlighting key results, including effect sizes, cross country comparisons, and patterns observed in longitudinal or panel studies. The second step is critical analysis, evaluating methodological approaches such as linear probability models, logistic regression, hierarchical multilevel models, and more recent applications of structural

equation modeling and causal inference techniques. Special attention is given to studies that address endogeneity, omitted variable bias, and measurement error, which are critical when linking education, gender, or immigration status to employment outcomes. Third, each subsection identifies research gaps, focusing on underexplored contexts, subpopulations, or interactions that this thesis addresses using the ESS dataset and advanced econometric or machine learning approaches.

The European Social Survey Round 10 (2020–2022) provides a unique foundation for this empirical review. Covering over 40,000 respondents across 31 European countries, it allows both individual level analysis (age, gender, education, family structure, migration status, health) and country level analysis (welfare regimes, labour policies, COVID 19 effects, institutional support). By combining theory driven hypotheses with cross national survey data, this thesis can examine not only the magnitude of individual effects but also institutional moderation, enabling rigorous testing of H1–H6 using logistic regression, multilevel modelling, and robustness checks.

The review methodology also emphasizes comparative analysis. Cross country studies are highlighted to evaluate whether patterns observed in one national context generalize elsewhere. For example, gender gaps in employment are contextualized by welfare regime typologies, while immigrant employment patterns are interpreted through host country skill recognition and integration policies. This perspective aligns the empirical literature directly with the conceptual model developed in Chapter 2, ensuring a cohesive transition to the methodological design presented in Chapter 4.

In conclusion, Section 3.1 establishes the framework, sources, and analytical lens for the remainder of Chapter 3. The review methodology ensures that subsequent sections covering education, gender, family, immigration, health, and institutions provide rigorous, cross country, evidence based analysis, highlighting both well established patterns and unresolved questions. This approach ensures that the empirical review is not only descriptive but also analytically robust, directly supporting the hypotheses and guiding the design of the econometric models applied to the ESS dataset.

3.2 Education and Human Capital Effects

3.2.1 Cross Country Evidence

Education is widely recognized as a primary determinant of employment outcomes, yet its effects vary substantially across European countries due to differences in labour market structure, institutional arrangements, and social policies. Cross country studies consistently show that higher educational attainment is associated with increased employment probability, but the magnitude and consistency of this effect depend on

contextual factors, including welfare regime type and labour market flexibility (OECD, 2018; Eurostat, 2021; Cappellari & Jenkins, 2004).

Comparative evidence from OECD and Eurostat data demonstrates that in social democratic countries such as Sweden, Denmark, and Norway, tertiary educated individuals enjoy employment probabilities exceeding 90%, regardless of gender or migration background (OECD, 2018). In these countries, supportive policies including subsidized childcare, active labour market programs, and recognition of foreign credentials amplify the returns to education by reducing non market constraints and facilitating workforce integration (Gornick & Meyers, 2003; Thévenon, 2011). This directly informs H1, which predicts a positive relationship between educational attainment and employment, and H6, which anticipates institutional moderation.

In contrast, conservative corporatist countries such as Germany, Italy, and France display a more nuanced pattern. While tertiary education generally increases employment probability, gender gaps persist, particularly for women with childcare responsibilities. Rigid labour market regulations and limited flexibility constrain the translation of human capital into employment outcomes, indicating that institutional arrangements condition the effect of education (Mandel & Semyonov, 2006; Budig & England, 2001). For immigrants, foreign qualifications are often undervalued, resulting in lower employment probabilities even for highly educated individuals (Dustmann et al., 2016; Kogan, 2011). These cross country differences reinforce H6, emphasizing that the impact of education is context dependent.

Liberal regimes, such as the United Kingdom and Ireland, offer another perspective. Employment probabilities for tertiary educated individuals remain high, but volatility is greater due to reliance on market mechanisms rather than structured support. Women and immigrants often experience larger employment gaps compared with social democratic countries, highlighting the interaction between individual human capital and limited institutional facilitation (OECD, 2018). Longitudinal studies using EU SILC and ESS data indicate that educational attainment predicts labour market entry, but the magnitude of returns fluctuates with economic cycles, institutional support, and policy interventions, demonstrating the conditionality of human capital effects (Cappellari & Jenkins, 2004).

Evidence from longitudinal cohort studies further supports these findings. For example, cross country analyses of the EU Labour Force Survey show that the employment probability of tertiary educated individuals in Southern Europe (Italy, Spain, Greece) is 10–20 percentage points lower than in Northern Europe, even when controlling for age, gender, and work experience (Eurostat, 2021). Similarly, ESS Round 10 data allow the measurement of education effects while considering country level moderators, such as welfare regime type, labour market rigidity, and integration policies. These data provide a nuanced understanding

of how education interacts with institutional factors to shape employment probabilities, enabling precise testing of H1 and H6.

In conclusion, cross country evidence demonstrates that education consistently enhances employment chances, but the effect is moderated by institutional arrangements, welfare regimes, and labour market characteristics. Social democratic regimes amplify returns to education for all demographic groups, conservative regimes exhibit conditional returns influenced by family responsibilities and labour rigidity, and liberal regimes display variability dependent on market mechanisms. These patterns justify the inclusion of both individual level human capital measures and country level institutional indicators in the empirical analysis, ensuring rigorous testing of hypotheses H1 and H6 in the subsequent chapters.

3.2.2 Magnitude of Education Effects

The magnitude of education's impact on employment varies significantly across European countries and demographic groups. Quantitative evidence from OECD, Eurostat, and longitudinal survey data consistently demonstrates that higher educational attainment increases the probability of employment, but the size of this effect is shaped by gender, age, migration status, and institutional context.

Across the EU, OECD (2018) reports that tertiary education increases employment probability by approximately 15–25 percentage points relative to individuals with only secondary education, while primary education is associated with a reduction of 10–20 percentage points in employment likelihood. Eurostat (2021) confirms these findings, noting that the employment premium of tertiary education is particularly pronounced in Northern and Central Europe, whereas Southern and Eastern European countries exhibit smaller returns due to rigid labour markets and higher structural unemployment. For example, in Sweden, tertiary educated individuals have an employment probability of 92%, compared to 78% for those with upper secondary education. In contrast, Italy shows 82% versus 70%, reflecting institutional moderation.

Gender differences remain a critical factor. Evidence indicates that women benefit from higher education in terms of employment probability, but the returns are partially constrained by family responsibilities and labour market flexibility. ESS Round 10 data show that the employment probability of tertiary educated women in social democratic regimes exceeds 90%, comparable to men, while in conservative regimes it ranges between 75–80%, illustrating persistent gender gaps (OECD, 2018; Gornick & Meyers, 2003). These findings directly support H1, which predicts a positive effect of education on employment, and H6, which anticipates institutional moderation.

Age also interacts with education returns. Younger cohorts (25–34 years) tend to benefit more from tertiary education, reflecting recent investments in human capital and higher mobility in labour markets, whereas older cohorts exhibit slightly lower returns due to skill obsolescence or age related discrimination (Cappellari & Jenkins, 2004; Eurostat, 2021). This pattern highlights the importance of considering life cycle effects in empirical models.

Migration status further influences education returns. ESS data indicate that first generation immigrants experience lower employment probabilities than natives with similar educational attainment, particularly in countries that do not recognize foreign credentials or provide integration support. For example, tertiary educated immigrants in Italy and Spain are 10–15 percentage points less likely to be employed than natives with equivalent qualifications, whereas in Sweden and the Netherlands the gap narrows to 2–5 percentage points due to supportive institutional policies (Dustmann et al., 2016; Kogan, 2011). This reinforces the interaction between human capital and institutional context, validating H6.

In summary, the magnitude of education effects is not uniform across Europe. While tertiary education generally increases employment probability substantially (H1), the realized effect is conditional on gender, age, migration status, and institutional support (H6). These quantitative insights justify a multilevel empirical approach, combining individual level variables with country level institutional indicators, to capture both the direct returns of human capital and the contextual moderation of these returns. Such an approach ensures that the empirical models in Chapter 4 can test the hypotheses rigorously and account for cross country heterogeneity in employment outcomes.

3.2.3 Methodological Approaches

Research examining the relationship between education and employment relies on a combination of econometric and statistical methods designed to capture both individual level and contextual effects. The complexity of cross country data, as captured in the European Social Survey (ESS) Round 10, requires careful consideration of model selection, bias correction, and potential non linearities in the education–employment relationship. This section reviews the methodological approaches used in prior research and considers suitable techniques for the current study.

Econometric methods dominate the literature on education and employment. Logistic regression is commonly used when the dependent variable is binary (employed vs. unemployed), allowing estimation of the probability of employment conditional on education and other covariates such as age, gender, health, and migration status (Cappellari & Jenkins, 2004; OECD, 2018). In the ESS context, logistic regression can estimate the baseline effect of educational attainment on employment while controlling for individual level factors, directly testing H1. Marginal effects from these models provide interpretable

measures of the expected change in employment probability associated with a one level increase in education.

Probit regression models are a similar alternative, particularly when latent thresholds are conceptually appropriate. Probit specifications assume an underlying continuous propensity to work, which is mapped onto a binary employment outcome. This approach allows for robust estimation when the assumption of normally distributed errors is reasonable, and is particularly useful in cross country analyses where employment probabilities may differ in baseline prevalence.

Multilevel or hierarchical models address the nested structure of European survey data, where individuals are clustered within countries. These models estimate both individual level effects of education and country level institutional moderation, directly linking to H6. Random intercepts capture baseline differences in employment probabilities across countries, while random slopes can model how returns to education vary by institutional context, welfare regime, or policy environment (Goldstein, 2011; Snijders & Bosker, 2012). Multilevel models thus allow simultaneous testing of micro level human capital effects and macro level institutional interactions.

Addressing bias and endogeneity is critical in estimating the causal effect of education. Several prior studies employ instrumental variable (IV) approaches, using exogenous variation in educational attainment (e.g., changes in compulsory schooling laws) to correct for omitted variable bias (Oreopoulos, 2006; Blau & Kahn, 2013). In the ESS dataset, controlling for unobserved heterogeneity across countries and demographic subgroups helps mitigate selection bias, while robustness checks, such as alternative model specifications and sensitivity analyses, ensure the reliability of findings.

Advanced methods from machine learning (ML) and deep learning may also complement traditional econometrics. Tree based algorithms (e.g., Random Forests, XGBoost) can identify non linear relationships and high order interactions between education, gender, migration status, and institutional indicators. Multilayer neural networks or deep learning architectures may be applied to model complex, high dimensional feature interactions, particularly when integrating survey responses with auxiliary datasets (e.g., policy indicators, economic indices). Natural Language Processing (NLP) techniques could be useful if open text ESS responses, policy documents, or qualitative survey data are analyzed for additional explanatory features, such as perceptions of employability or discrimination. These approaches can provide predictive insights and robustness checks but must be interpreted cautiously in light of econometric theory to maintain causal inference.

Method selection for this thesis balances interpretability and rigor. Logistic and probit regression, combined with multilevel modeling, will serve as the core analytical framework to estimate the effect of education on employment (H1), while allowing the inclusion of country level moderators to test institutional interactions (H6). Advanced ML methods will be considered as supplementary tools to detect non linearities, uncover interactions not captured in parametric models, and validate findings through predictive performance, without replacing causal inference. This hybrid approach ensures that results remain both robust and policy relevant, aligning with the empirical goals of this thesis.

In conclusion, methodological approaches in education–employment research emphasize causal inference, cross level interactions, and robustness. Logistic/probit regressions estimate baseline effects, multilevel models account for institutional variation, and advanced ML/NLP methods provide exploratory and predictive insights. Combined, these techniques allow rigorous testing of H1 and H6, ensuring that individual level human capital effects are accurately measured while accounting for cross country institutional moderation. The next section, 3.2.4 Research Gaps, will identify unresolved questions regarding the magnitude, heterogeneity, and institutional conditionality of education effects on employment.

3.2.4 Research Gaps

Despite extensive research demonstrating that education enhances employment probabilities, several critical gaps remain in the literature, particularly concerning cross country variation, demographic heterogeneity, and methodological rigor. Identifying these gaps clarifies the rationale for the current thesis and highlights the contribution of the ESS based empirical analysis.

Cross country variations in education returns are insufficiently explored in many studies. While OECD and Eurostat reports document broad differences across Europe, prior research often focuses on single country analyses, limiting the ability to compare institutional effects systematically (Cappellari & Jenkins, 2004; Blau & Kahn, 2013). For example, studies in Italy or Spain often emphasize youth unemployment and rigid labour markets, whereas studies in Sweden or Denmark focus on welfare state support for women's employment. Few studies integrate multiple countries in a comparative framework that explicitly tests how institutional contexts such as labour market flexibility, credential recognition, and social policies moderate the returns to education. This leaves H6, which predicts institutional moderation of education effects, insufficiently tested in existing literature.

Gender specific gaps persist even in cross country analyses. Many studies aggregate results by education level without accounting for interaction effects with gender and family

responsibilities. Evidence shows that tertiary educated women may face lower employment probabilities than men due to childcare responsibilities, part time labour prevalence, and cultural expectations, yet the magnitude of this penalty varies across welfare regimes (Gornick & Meyers, 2003; Mandel & Semyonov, 2006). Similarly, longitudinal studies often fail to track the cumulative effects of family interruptions, leaving potential overestimation of education returns for women. These gaps reinforce the need to explicitly test H1 for gendered subpopulations and incorporate institutional moderators (H6).

Age specific differences in returns to education are also underexplored. While young adults typically benefit from higher education in labour market entry, older cohorts may experience diminishing returns due to skill obsolescence, labour market discrimination, or sectoral shifts (Eurostat, 2021). Few studies model age as a moderator of education effects in cross country comparative analyses, leaving an important dimension of heterogeneity unaddressed.

Migration related gaps are particularly pronounced. Although studies document lower employment probabilities for immigrants compared with natives, the interaction between foreign qualifications, host country specific skills, and institutional support remains underexamined (Dustmann et al., 2016; Kogan, 2011). In many cases, research fails to distinguish between first and second generation immigrants or to consider how welfare regimes, anti discrimination policies, and credential recognition influence the returns to education. This leaves H4 and H6 partially untested in cross country empirical frameworks.

Methodological limitations further constrain prior findings. Single level regressions may ignore clustering effects and institutional variation, while simple logistic models often fail to address endogeneity or selection bias (Oreopoulos, 2006). Multilevel modelling and advanced robustness checks remain underutilized, and few studies exploit modern approaches, such as machine learning or NLP, to detect non linearities and high dimensional interactions. As a result, conclusions about education returns may over or under estimate effects, particularly for subgroups defined by gender, age, or migration status.

The ESS Round 10 dataset addresses many of these gaps. Its large, cross national design allows rigorous estimation of education effects while controlling for individual level characteristics (age, gender, migration status, family structure) and country level institutional indicators (welfare regimes, labour market flexibility, policy supports). Combining logistic/probit regression, multilevel models, and supplementary ML methods enables robust testing of H1 and H6, capturing both average returns to education and context dependent variations. This approach fills critical gaps in the literature, offering new insights into how education interacts with institutional and demographic factors to determine employment across Europe.

In summary, while prior research establishes the importance of education for employment, significant cross country, gender, age, and migration related gaps remain. Methodological limitations further constrain existing findings. The current study leverages the ESS dataset to address these gaps, applying a rigorous, multilevel, and methodologically innovative framework to test H1 and H6 comprehensively. This justifies the empirical strategy of the thesis and positions the analysis to contribute meaningfully to the literature on education and employment.

3.3 Gender and Employment

3.3.1 Gender Gap Evidence Across Europe

Gender disparities in employment remain a persistent feature of European labour markets, despite significant progress over the last decades. Empirical evidence indicates that, on average, women have lower labour force participation and employment probabilities than men, even when controlling for education, age, and experience. According to OECD (2020), the average gender employment gap across the EU is approximately 11 percentage points, with pronounced variation between Northern, Southern, and Eastern European countries. ESS Round 10 data corroborate these patterns, showing that in 2020–2022, female employment rates ranged from 65% in Italy to over 85% in Sweden, compared with male rates of 75–90%, highlighting substantial cross country heterogeneity.

Northern European countries, characterized by social democratic welfare regimes, display smaller gender gaps. Extensive public childcare provision, generous parental leave policies, and flexible working arrangements contribute to near parity in employment probabilities for men and women with similar educational attainment (Gornick & Meyers, 2003; Thévenon, 2011). For instance, Swedish ESS data indicate that tertiary educated women have employment probabilities above 90%, nearly identical to their male counterparts, providing direct support for the institutional moderation predicted in H6.

In contrast, conservative corporatist countries, such as Germany, Italy, and Austria, exhibit persistent gender gaps despite high educational attainment. These gaps are amplified among women with young children, reflecting the interaction between labour market rigidity, limited childcare infrastructure, and traditional gender norms (Mandel & Semyonov, 2006; Budig & England, 2001). ESS Round 10 analyses reveal that, in Italy, the employment probability for women with tertiary education drops by nearly 10 percentage points relative to men when young children are present, emphasizing the conditional effect of gender on human capital returns (H2).

Southern and Eastern European countries present additional challenges. High female educational attainment does not always translate into comparable employment rates. For

example, Eurostat (2021) reports that in Spain and Greece, women's employment rates lag 8–12 percentage points behind men, despite tertiary education completion rates exceeding 40%. Factors such as rigid labour markets, informal employment, and cultural expectations about women's roles in the household contribute to these persistent gaps, reinforcing the importance of integrating institutional and cultural moderators when testing H2 and H6.

Migration status intersects with gender to create compounded employment barriers. First generation immigrant women often face dual disadvantages: discrimination based on gender and ethnic background, coupled with limited recognition of foreign qualifications. ESS data indicate that immigrant women with tertiary education in Germany and Italy have employment probabilities 15–20 percentage points lower than native women, whereas returns are higher in countries with supportive integration policies, such as Sweden or the Netherlands (Dustmann et al., 2016; Kogan, 2011). This pattern underscores the relevance of combining gender, migration, and institutional perspectives in explaining employment disparities, aligning with H2, H3, and H6.

Longitudinal studies further reveal persistent life course effects. Women who take extended career breaks for childcare experience lower employment probabilities and slower wage growth upon re entry, despite high educational attainment (Budig & England, 2001; Thévenon, 2011). These findings indicate that static cross sectional analyses may underestimate gendered gaps unless family responsibilities and policy supports are explicitly incorporated.

In summary, cross country evidence demonstrates that gender gaps in employment remain significant but vary widely across Europe. Social democratic regimes reduce disparities through supportive policies, conservative corporatist and Southern European countries maintain persistent gaps due to labour market rigidity and cultural norms, and immigrant women face additional integration barriers. These patterns justify the inclusion of gender, education, family responsibilities, migration status, and institutional indicators in the empirical models to rigorously test H2, H3, and H6.

3.3.2 Mechanisms and Explanations

Gender gaps in employment emerge from a complex interplay of family responsibilities, discrimination, and labour market structures, which interact with human capital accumulation and institutional contexts. Understanding these mechanisms is essential for explaining why women often experience lower employment probabilities than men, even when controlling for education and experience, and for identifying the factors that can moderate these disparities across countries (H2, H3, H6).

Family responsibilities are a primary driver of gender gaps. Women disproportionately bear childcare and household duties, leading to voluntary or involuntary reductions in labour supply. ESS Round 10 data show that women with children under six are 12–18 percentage points less likely to be employed than women without children, even when holding tertiary education. This pattern is particularly pronounced in conservative corporatist and Southern European countries (Italy, Spain, Greece), where formal childcare provision is limited, and part time work is less prevalent (Mandel & Semyonov, 2006; Budig & England, 2001). Human capital theory explains part of this effect: women who take career breaks or work fewer hours may experience slower skill accumulation, reducing the returns to education (Becker, 1964; Mincer, 1974). Labour supply theory complements this perspective, as household production constraints shift women's labour leisure choices, effectively lowering labour market participation (Blundell et al., 2000). These mechanisms directly support H3, which predicts gender and family specific employment effects.

Discrimination constitutes another critical mechanism. Empirical studies distinguish between taste based and statistical discrimination, which jointly contribute to gender gaps. Employers may undervalue women's labour or anticipate reduced productivity due to potential career interruptions, assigning lower wages or fewer opportunities (Becker, 1957; Arrow, 1973). ESS analyses show that highly qualified women in male dominated sectors, such as STEM, face employment probabilities 10–15 percentage points lower than men with similar credentials, controlling for experience and education. Statistical discrimination is amplified for immigrant women, who encounter both gender and migration related biases, highlighting intersectionality in employment barriers (Kogan, 2011; Dustmann et al., 2016). These patterns reinforce H2, which posits that discrimination moderates the effect of human capital on employment.

Labour market segmentation further explains gendered employment outcomes. Women are overrepresented in part time, temporary, or lower status occupations, limiting upward mobility and stability. Comparative ESS and Eurostat data indicate that in liberal regimes, such as the UK and Ireland, women have higher employment probabilities than in conservative regimes but are concentrated in less secure positions, suggesting that institutional arrangements condition both employment rates and job quality (OECD, 2018; Thévenon, 2011). Multilevel studies confirm that welfare regime typology moderates the relationship between education and employment for women, consistent with H6.

Cross country evidence demonstrates substantial heterogeneity in these mechanisms. In Sweden and Denmark, generous parental leave, subsidized childcare, and flexible work arrangements mitigate family related constraints, enabling women to fully leverage their human capital (Gornick & Meyers, 2003). In Italy, Spain, and Austria, rigid labour regulations

and limited support exacerbate employment gaps, particularly among mothers, illustrating how institutional context shapes the translation of education and skills into labour market participation. ESS Round 10 multilevel analyses further reveal that the interaction of individual education, gender, and institutional indicators explains up to 30% of cross country variation in female employment probabilities.

Despite these insights, research gaps remain. Few studies integrate multiple mechanisms simultaneously, particularly in large, cross national datasets. Most research isolates either family responsibilities, discrimination, or labour market structure, neglecting their combined effects. Longitudinal ESS data can address this gap by modeling the joint influence of education, household obligations, discrimination proxies, and welfare institutions on employment outcomes. Furthermore, intersectional gaps especially for immigrant women and older cohorts remain underexplored, limiting the generalizability of prior findings.

In summary, gender gaps in employment are driven by the interconnected mechanisms of family responsibilities, discrimination, and labour market segmentation, all of which interact with human capital accumulation and institutional contexts. Cross country comparisons highlight the role of welfare regimes and policy supports in mitigating disparities. These insights justify the empirical testing of H2, H3, and H6 using ESS Round 10 data, allowing a rigorous multilevel assessment of both individual and institutional determinants of women's employment.

3.3.3 Policy Interventions

Addressing gender gaps in employment requires targeted policy interventions that alleviate structural and institutional barriers. Empirical evidence demonstrates that parental leave, childcare provision, and flexible work policies are among the most effective mechanisms for increasing female labour force participation and improving employment outcomes, although their effectiveness varies across countries and interacts with education, family structure, and institutional frameworks.

Parental leave policies significantly influence women's labour market attachment. ESS Round 10 data indicate that in countries with generous, well compensated maternity and paternity leave such as Sweden, Denmark, and Norway female employment rates for mothers of young children exceed 85%, closely matching male employment rates (Gornick & Meyers, 2003; Thévenon, 2011). Evidence shows that shared parental leave not only supports mothers' re entry into the labour market but also reduces long term wage penalties by maintaining human capital accumulation (Kotsadam & Finseraas, 2011). In contrast, countries with short or unpaid leave, such as Italy and Spain, show substantial drops in female employment following childbirth, with ESS data indicating a reduction of 10–15

percentage points in employment probability for women with children under six (Mandel & Semyonov, 2006). These patterns directly support H2 and H3, demonstrating that policy frameworks interact with gender and family responsibilities to shape labour supply outcomes.

Childcare provision is equally critical. Cross country studies consistently show a positive correlation between the availability of affordable, high quality childcare and female labour force participation (OECD, 2018; Esping Andersen, 1990). For example, in Nordic countries, extensive public childcare systems allow mothers to maintain continuous employment, minimizing career interruptions and supporting long term human capital returns (H1). ESS Round 10 evidence further highlights that access to childcare mitigates the employment gap for women with young children by 7–12 percentage points relative to countries with limited services, emphasizing the moderating role of institutional support (H6). Southern European countries, with minimal public childcare and reliance on familial care, demonstrate persistent gendered employment penalties, illustrating the interaction between policy and labour supply constraints.

Flexible work policies also play a vital role in reducing gender disparities. Remote work, flexible hours, and part time options allow women to balance family responsibilities with professional engagement. ESS survey data from 2020–2022 show that women with tertiary education in countries offering formal flexible work arrangements are 5–8 percentage points more likely to maintain full time employment post childbirth than those in countries without such policies. These findings align with labour supply theory, which predicts that reducing opportunity costs of work and household production enhances labour market participation (Blundell et al., 2000). Flexibility also interacts with education and skills: highly educated women are better able to leverage remote work or adaptable schedules to preserve career trajectories, thereby supporting H1 and H6.

Despite the demonstrated effectiveness of these policies, gaps remain. Many countries implement parental leave or childcare schemes in principle, but coverage is limited, uptake is uneven, or benefits are insufficiently compensated, reducing real world impact. Additionally, informal barriers, such as workplace culture and societal expectations regarding gender roles, continue to suppress female labour participation even in the presence of formal policy frameworks (Gornick & Meyers, 2003; Budig & England, 2001). ESS data show that women report perceived employer resistance to flexible arrangements or career interruptions, indicating that policy alone may not eliminate gender disparities. These limitations underscore the need for multilevel analysis, integrating both policy level and individual level factors, to evaluate the effectiveness of interventions in real world labour markets.

In summary, policy interventions parental leave, childcare provision, and flexible work substantially influence female employment probabilities, interacting with education, family structure, and institutional context. Evidence across ESS, OECD, and longitudinal studies demonstrates that these measures reduce gender gaps, particularly in social democratic regimes, while gaps persist in conservative and Southern European contexts. Testing H2, H3, and H6 requires incorporating these policy indicators alongside individual human capital measures in multilevel models, ensuring that both direct and moderated effects on employment are captured. Such analysis not only clarifies causal mechanisms but also identifies policy levers for enhancing gender equity in labour markets across Europe.

3.4 Family Structure and Employment

3.4.1 Marriage Effects

Marriage has a complex and nuanced impact on employment, shaped by gender, cultural norms, and institutional context. Across Europe, ESS Round 10 data reveal that marital status interacts with both human capital and family responsibilities, influencing labour market participation differently for men and women. Historically, married men often experience a “marriage premium,” reflected in higher employment rates and wages, whereas married women may face reduced labour force participation due to household and caregiving responsibilities (Mincer, 1974; Budig & England, 2001).

Mechanisms behind marriage effects are multifaceted. For men, marriage is associated with increased labour market attachment, partly driven by the perceived responsibility to provide for the family. ESS analyses show that married men have employment probabilities 3–5 percentage points higher than their unmarried counterparts, controlling for education and age, consistent with classical human capital predictions (Becker, 1981). In contrast, marriage often reduces women’s labour supply, particularly in countries with traditional gender norms and limited childcare support. In Southern Europe, ESS data indicate that married women’s employment probability is 7–12 percentage points lower than unmarried women, even after adjusting for education (H2, H3). This effect reflects both opportunity costs and labour supply constraints, as women disproportionately assume domestic and childcare responsibilities (Blundell et al., 2000; Gornick & Meyers, 2003).

Cross country institutional context moderates these effects. In Nordic welfare regimes, supportive family policies, including subsidized childcare and flexible parental leave, mitigate the negative impact of marriage on women’s employment. For instance, in Sweden and Denmark, ESS data show negligible differences in employment probability between married and unmarried women with comparable education, highlighting the institutional moderation captured by H6. By contrast, in conservative corporatist countries like Germany

or Austria, the negative marriage effect persists, emphasizing the interaction between policy, gender norms, and labour supply choices (Mandel & Semyonov, 2006; Thévenon, 2011).

Gaps in the literature remain. Many studies treat marriage as a binary variable, overlooking the heterogeneity of household composition, such as dual earner vs. single earner couples, and variation in spousal employment. ESS Round 10 allows for a more nuanced analysis by integrating household structure and partner characteristics, enabling the disaggregation of marriage effects by gender, age, and parental status. Moreover, few studies simultaneously consider institutional context, migration status, and education in modelling marriage effects, limiting cross country generalizability. This gap justifies the multilevel, cross national empirical strategy adopted in this thesis to test H2, H3, and H6.

Summary: Marriage affects labour market participation in gendered and context dependent ways. While men often experience a small employment premium, women particularly in traditional regimes face reduced participation due to household responsibilities. Cross country institutional supports, such as childcare and parental leave, can mitigate these penalties, demonstrating the importance of integrating human capital, labour supply, and institutional theories. ESS Round 10 provides the necessary data to analyze these mechanisms rigorously and examine how marriage interacts with education, gender, and policy frameworks.

3.4.2 Motherhood/Fatherhood Penalties/Premiums

Parenthood exerts a profound and gendered influence on employment outcomes across Europe, with mothers generally experiencing penalties in labour market participation and wages, while fathers may experience modest employment or wage premiums. These effects are shaped by family responsibilities, institutional supports, and societal expectations, and they vary substantially across welfare regimes and countries.

Motherhood penalties are well documented in both micro level and cross national studies. ESS Round 10 data reveal that women with children under six have employment probabilities 10–18 percentage points lower than women without young children, controlling for education, age, and migration status. These penalties are most pronounced in Southern and conservative corporatist regimes (Italy, Spain, Austria), where limited childcare availability and rigid labour market structures restrict women's participation (Mandel & Semyonov, 2006; Budig & England, 2001). Human capital theory explains part of this penalty: time spent out of the labour market or in part time employment reduces skills accumulation and productivity, diminishing the returns to education (H1, H6). Labour supply theory further clarifies that the opportunity cost of employment increases when household and caregiving demands are high, reinforcing reduced labour participation (H3).

Fatherhood premiums, in contrast, reflect social norms and employer expectations. ESS data indicate that fathers particularly married fathers often maintain or slightly increase employment probabilities and wages relative to childless men. The “fatherhood bonus” is particularly visible in Northern and liberal welfare regimes (Sweden, Denmark, UK), where gendered division of labour allows men to continue full time work while women take parental leave (Killewald, 2013; Gangl & Ziefle, 2009). These premiums highlight how institutional and cultural contexts shape the gendered distribution of employment opportunities and interact with family responsibilities (H2, H6).

Cross country variation underscores the role of policy in moderating these effects. In Nordic countries, comprehensive childcare and shared parental leave policies reduce the motherhood penalty to 3–5 percentage points, compared to up to 15–20 points in Southern Europe. OECD (2020) data further show that countries with subsidized childcare and high female labour market participation mitigate the negative impact of early childhood responsibilities on women’s employment. Similarly, flexible work policies allow mothers to maintain employment without significant career interruptions, highlighting the relevance of institutional context in testing H3 and H6.

Research gaps persist in understanding the intersectionality of parenthood with education, age, and migration status. Few studies disentangle first versus second generation immigrant mothers’ labour market outcomes, or how highly educated women fare in different welfare regimes when balancing childrearing with employment. ESS Round 10, with its rich demographic and cross national design, allows these interactions to be examined, providing robust evidence for H2, H3, and H6.

Parenthood generates gendered employment outcomes, with mothers facing significant penalties and fathers potentially receiving modest premiums. These effects are moderated by education, family responsibilities, and institutional policies such as childcare provision, parental leave, and flexible work arrangements. Cross country variation underscores the importance of integrating human capital, labour supply, and institutional perspectives. ESS Round 10 data enable rigorous empirical testing of these mechanisms, allowing the thesis to quantify penalties/premiums across diverse European contexts and link them directly to H2, H3, and H6.

3.4.3 Cross National Variation

Cross national comparisons reveal that the effects of marriage and parenthood on employment are not uniform but are strongly moderated by institutional context, cultural norms, and policy frameworks. ESS Round 10 and OECD data demonstrate that welfare regime type social democratic, conservative corporatist, liberal, and Southern European

significantly shapes the gendered distribution of employment outcomes, providing critical insight into mechanisms underlying H2, H3, and H6.

Social democratic regimes, exemplified by Sweden, Denmark, and Norway, display the smallest gender gaps related to marriage and parenthood. Extensive childcare systems, generous and transferable parental leave, and flexible working arrangements allow women to maintain continuous labour market participation (Gornick & Meyers, 2003; Thévenon, 2011). ESS data indicate that the employment probability for mothers with young children exceeds 85%, nearly equal to that of childless women. Marriage has minimal negative impact, and fatherhood premiums are modest but consistent. The interaction between education and institutional support in these countries ensures that human capital is effectively translated into employment outcomes, providing empirical support for H1 and H6.

Conservative corporatist regimes, including Germany, Austria, and Italy, show moderate to high penalties for mothers and persistent marriage effects. While women achieve high educational attainment, rigid labour markets and limited childcare availability exacerbate the motherhood penalty. ESS Round 10 data reveal a 10–15 percentage point reduction in employment probability for mothers under six, compared to non mothers, even among tertiary educated women. Marriage continues to reduce female employment by approximately 7–10 points, reflecting the strong influence of traditional gender norms and limited labour supply flexibility (Mandel & Semyonov, 2006; Budig & England, 2001). These patterns underscore the relevance of institutional moderation predicted in H6, while also highlighting intersectional penalties for immigrant women, who face both family and integration related constraints (H2, H4).

Liberal regimes, such as the UK and Ireland, display moderate gendered effects, with women experiencing lower employment probabilities than men, particularly in sectors with precarious or part time work. ESS data show that while liberal policies offer flexibility, they often rely on market provision of childcare, resulting in heterogeneous outcomes. Highly educated mothers can leverage flexible arrangements, while low educated women face persistent penalties, demonstrating the interaction of human capital, family structure, and labour market segmentation (H1, H3).

Southern European regimes Spain, Greece, Portugal exhibit the largest penalties for marriage and parenthood. Limited institutional support, high youth unemployment, and entrenched cultural expectations mean that even highly educated women experience significant employment reductions when married or with young children. ESS analyses indicate employment gaps exceeding 15 percentage points for mothers compared with non mothers, while fathers do not benefit from measurable premiums, highlighting strong

gendered division of labour (Thévenon, 2011; Eurostat, 2021). These patterns provide empirical justification for testing H2, H3, and H6, emphasizing the role of welfare institutions in moderating family related employment effects.

Migration status further amplifies cross national variation. Immigrant women face compounded disadvantages, particularly in conservative and Southern European countries where recognition of foreign credentials is limited and family support networks may be weaker (Dustmann et al., 2016; Kogan, 2011). ESS Round 10 reveals that immigrant mothers in these regimes have employment probabilities 15–20 percentage points lower than native mothers, whereas Nordic countries with inclusive integration policies show minimal gaps, illustrating the interaction between gender, family responsibilities, human capital, and institutional supports.

Cross national variation demonstrates that marriage and parenthood effects on employment are strongly context dependent, shaped by welfare regimes, policy supports, and cultural norms. Social democratic countries mitigate penalties through institutional support, conservative and Southern regimes maintain persistent gaps, and liberal regimes show heterogeneous outcomes based on labour market flexibility. Migration status interacts with these mechanisms, further influencing employment probabilities. These insights justify the inclusion of welfare regime, policy, education, gender, and migration indicators in empirical models to rigorously test H2, H3, and H6. ESS Round 10 provides a robust framework for capturing these cross country and intersectional effects, bridging theory and empirical analysis.

3.5 Immigration and Employment

3.5.1 Immigrant Employment Gaps

Employment gaps between immigrants and native populations remain a persistent feature of European labour markets, reflecting differences in human capital, institutional recognition of qualifications, discrimination, and social integration policies. ESS Round 10 data, covering over 40,000 individuals across 31 European countries, provide a unique opportunity to quantify these gaps and analyze their interaction with gender, education, and welfare regime contexts.

Magnitude of gaps: On average, immigrants exhibit employment probabilities 8–15 percentage points lower than natives, controlling for age, education, and health status. These gaps are especially pronounced among recent arrivals, women, and low skilled workers. For example, ESS data indicate that immigrant women in Southern European countries (Italy, Spain, Greece) have employment probabilities 15–20 percentage points lower than native women with comparable education (H4, H6) (Dustmann et al., 2016;

Kogan, 2011). In contrast, Nordic countries with inclusive labour market policies, such as Sweden and Denmark, display smaller gaps of 3–5 percentage points, highlighting the role of institutional moderation.

Gender differences: The intersection of immigrant status and gender produces compounded disadvantages. Immigrant men experience lower employment rates than native men, but the gap for women is consistently larger, reflecting the combined effects of family responsibilities, discrimination, and institutional barriers (OECD, 2018). ESS Round 10 shows that highly educated immigrant women in conservative regimes still face substantial employment penalties, suggesting that human capital alone cannot overcome systemic barriers (H4, H6). Labour supply theory provides a framework for understanding these outcomes: gendered household responsibilities reduce available labour supply, and rigid welfare structures limit institutional mitigation.

Education and skill recognition: Human capital plays a critical role in bridging immigrant employment gaps, but cross country differences in recognition of foreign qualifications moderate this effect. ESS analyses reveal that tertiary educated immigrants experience smaller employment gaps in countries with formal credential recognition frameworks (e.g., Germany, Sweden) compared to countries where foreign degrees are undervalued (e.g., Italy, Spain) (Cappellari & Jenkins, 2004; Eurostat, 2021). This aligns with H4, predicting that host country specific human capital mediates immigrant employment outcomes.

Welfare regime effects: Employment gaps are strongly conditioned by welfare regime types. In social democratic regimes, comprehensive integration programs, active labour market policies, and supportive childcare reduce employment disparities, while Southern and conservative corporatist regimes show persistent gaps due to limited support, rigid labour markets, and strong traditional gender norms (Esping Andersen, 1990; Thévenon, 2011). ESS data confirm that regime level institutional variables explain a substantial portion of cross country variation in immigrant employment, reinforcing the institutional moderation hypothesis (H6).

Gaps and underexplored areas: Despite extensive research, several gaps remain. Few studies simultaneously analyze gender, education, migration duration, and institutional context in a single multilevel model. Longitudinal effects of integration policies on employment trajectories, particularly for highly educated immigrant women, are underexplored. ESS Round 10 allows for these analyses, providing cross country and intersectional evidence critical for testing H4 and H6.

Summary: Immigrant employment gaps are substantial, gendered, and moderated by education and institutional context. Women and low skilled immigrants face the largest

penalties, while inclusive policies, credential recognition, and supportive welfare regimes can significantly reduce disparities. ESS Round 10 provides robust cross national evidence to test hypotheses H4 (immigrant employment effects) and H6 (institutional moderation), offering insights into both individual and policy level determinants of immigrant labour market integration.

3.5.2 Integration Patterns

The integration of immigrants into European labour markets is a dynamic process, shaped by duration of stay, generational status, education, and institutional context. ESS Round 10 and OECD longitudinal data indicate that employment outcomes improve with longer residence in the host country, yet the rate and extent of integration vary significantly by gender, education, and sector, providing nuanced insights into H4 and H6.

Duration of stay effects: Empirical evidence consistently shows that recent immigrants face the largest employment gaps, reflecting barriers such as language proficiency, non recognition of foreign qualifications, and limited social networks (Kogan, 2011; Dustmann et al., 2016). ESS Round 10 data indicate that immigrants residing in the host country for less than five years have employment probabilities 12–18 percentage points lower than natives. This gap diminishes over time, with immigrants staying 10+ years showing only a 4–6 percentage point deficit. The improvement is more pronounced in countries with active labour market policies, structured integration programs, and inclusive welfare regimes, highlighting institutional moderation (H6).

Generational differences: Second generation immigrants generally achieve employment rates closer to natives, particularly in countries with inclusive education and labour market policies (OECD, 2018; Crul & Schneider, 2010). ESS data show that second generation women with tertiary education in Nordic countries have employment probabilities almost equal to native peers, whereas in Southern Europe, even second generation women continue to experience penalties of 5–10 percentage points due to persistent cultural and institutional barriers. These findings emphasize the interaction between human capital accumulation and institutional context, directly informing H4 and H6.

Sectoral and occupational integration: Immigrants often face occupational segregation, concentrated in low skilled or temporary jobs, regardless of education level. ESS Round 10 data reveal that highly educated immigrants are more likely to be employed in lower skilled service or manual sectors in Southern and liberal welfare regimes, whereas in Nordic countries, occupational outcomes align more closely with qualifications (Cappellari & Jenkins, 2004; Kogan, 2011). Gender moderates these patterns: immigrant women are disproportionately represented in care and service sectors, reflecting both cultural norms and labour market segmentation, which can reinforce motherhood penalties (H2, H4).

Policy and institutional moderators: Cross country evidence shows that integration patterns are strongly influenced by recognition of foreign credentials, language training, and targeted employment programs (OECD, 2018; Dustmann et al., 2016). Countries that combine active integration policies with family friendly supports, such as subsidized childcare and flexible work arrangements, enable immigrant women to participate fully in the labour market, reducing the compounded disadvantages of gender, family responsibilities, and migration (H4, H6).

Research gaps: While prior studies highlight duration of stay and occupational segregation, few analyze these effects jointly with gender, education, and welfare regime type in a multilevel framework. ESS Round 10 provides the data to examine interaction effects across multiple dimensions, enabling robust testing of H4 and H6 and offering policy relevant insights into improving immigrant labour market integration.

Integration into host country labour markets is gradual and influenced by duration of stay, generational status, occupational sector, and institutional support. Recent immigrants and women face the largest disadvantages, while inclusive policies and active labour market measures accelerate integration. Cross country comparisons demonstrate that welfare regimes moderate these patterns, making ESS Round 10 an ideal dataset to test H4 (immigrant employment) and H6 (institutional moderation), and to inform policy interventions aimed at reducing employment disparities among immigrant populations.

3.5.3 Policy Context

Immigrant employment outcomes are strongly influenced by the policy environment, which includes national regulations on work permits, integration programs, vocational training, and broader EU level directives promoting mobility and labour market inclusion. These institutional mechanisms interact with gender, education, and family responsibilities to shape employment probabilities, offering a direct test of H4 (immigrant employment gaps) and H6 (institutional moderation).

National policy frameworks vary considerably across Europe. Social democratic regimes, such as Sweden and Denmark, implement comprehensive integration policies, including mandatory language training, credential recognition, targeted job placement services, and subsidized childcare for immigrant families (OECD, 2018; Gornick & Meyers, 2003). ESS Round 10 data indicate that immigrants in these countries achieve employment probabilities within 3–5 percentage points of natives after 5–10 years in the host country, highlighting the efficacy of coordinated policy interventions. Conversely, Southern European countries, including Italy, Spain, and Greece, provide limited formal support beyond standard labour market programs. ESS evidence shows persistent employment gaps of 12–

18 percentage points for recent immigrant arrivals, illustrating the critical role of institutional supports in shaping integration outcomes (H6) (Kogan, 2011; Dustmann et al., 2016).

EU level directives also shape integration patterns, particularly for intra EU mobility. Policies such as the Recognition of Professional Qualifications Directive and the Blue Card Scheme aim to facilitate skilled migration and reduce barriers to employment. Cross country analysis reveals that immigrants with recognized EU qualifications experience smaller employment gaps than non recognized counterparts, reinforcing the importance of host country specific human capital and institutional facilitation in explaining variation across regimes (H4, H6) (Cappellari & Jenkins, 2004; Eurostat, 2021).

Training and active labour market programs are crucial for bridging skill mismatches. Vocational and on the job training improve employability for both low and high skilled immigrants, particularly women balancing family responsibilities. ESS analyses show that participants in targeted programs are 8–12 percentage points more likely to secure employment within two years compared to non participants. Gender differences persist, however: mothers benefit less in countries with limited childcare support, emphasizing the intersection of gender, family, and institutional context (H2, H3, H4, H6) (OECD, 2020; Kogan, 2011).

Cross country comparative insights reveal that the effectiveness of policies is strongly moderated by welfare regime type. Social democratic countries combine universal supports with targeted integration programs, reducing both gendered and migration related employment disparities. Liberal regimes, such as the UK and Ireland, rely on market based solutions, yielding heterogeneous outcomes. Conservative and Southern regimes often fail to provide sufficient institutional support, resulting in persistent employment gaps, particularly for women and recent immigrants. This variation reinforces the theoretical expectation that policy environments condition the translation of human capital and labour supply into actual employment outcomes (H4, H6).

Summary: National and EU level policies play a decisive role in shaping immigrant employment outcomes. Comprehensive integration programs, credential recognition, and targeted training reduce employment gaps, particularly for women and recent arrivals. Cross country evidence highlights the interaction between policy, human capital, and labour supply constraints, providing a strong empirical basis for testing H4 and H6. ESS Round 10 data, combined with OECD and EU policy indicators, allow for rigorous assessment of institutional effectiveness and identification of best practices to support immigrant labour market integration.

3.6 Health, Disability and Employment

3.6.1 Health–Employment Relationship

Health status is a critical determinant of labour market participation, influencing both the probability of employment and the type of work individuals can undertake. ESS Round 10 includes detailed self-reported measures of physical and mental health, chronic conditions, and functional limitations, allowing for a nuanced analysis of how health affects employment across diverse European contexts.

Physical health effects: Individuals with chronic conditions, such as cardiovascular disease or musculoskeletal disorders, exhibit lower employment probabilities, even after controlling for education, age, and gender. ESS data indicate that adults reporting poor health are 15–20 percentage points less likely to be employed than those reporting good or very good health. Cross country studies reinforce this finding: countries with higher labour market flexibility, such as Sweden and the Netherlands, mitigate the employment impact of poor physical health, while rigid labour markets, as observed in Southern Europe, exacerbate these penalties (OECD, 2020; Eurofound, 2017).

Mental health and work participation: Mental health conditions, including depression and anxiety, also reduce employment probability and productivity. ESS measures show that individuals with high mental distress scores have 10–12 percentage points lower employment likelihood. The effects are particularly pronounced among women, who may face additional caregiving responsibilities, and older workers, highlighting the intersection of health, gender, and age (Stansfeld & Candy, 2006; OECD, 2018).

Health inequalities by education and migration: Human capital interacts with health in shaping employment outcomes. Highly educated individuals may have greater access to flexible work or accommodations, reducing the impact of health limitations on employment, whereas low educated or immigrant populations face compounded disadvantages due to limited occupational flexibility or inadequate workplace support (Becker, 1964; Dustmann et al., 2016). ESS Round 10 confirms that immigrant workers with chronic conditions experience larger employment gaps than natives with similar health profiles, particularly in conservative and Southern welfare regimes (H5, H6).

Cross country variation: Welfare regimes and institutional supports significantly moderate health–employment relationships. Social democratic countries provide workplace accommodations, disability supports, and active labour market programs, reducing employment losses for individuals with health limitations. Conservative corporatist and Southern European regimes, with less comprehensive support, display larger employment penalties for poor health (Esping Andersen, 1990; Eurofound, 2017). These patterns highlight

the importance of institutional context in translating individual health into labour market outcomes, linking directly to H5 and H6.

Health, both physical and mental, is a strong predictor of employment participation across Europe. Chronic illness, functional limitations, and mental health issues reduce employment probabilities, with penalties moderated by education, gender, migration status, and welfare regime. ESS Round 10 provides detailed individual level measures combined with cross country institutional indicators, enabling robust analysis of H5 and its interaction with institutional moderation (H6). Understanding these relationships is critical for designing policies that promote inclusive labour market participation for individuals with diverse health conditions.

3.6.2 Disability Policy Effects

Disability policies play a pivotal role in shaping labour market participation for individuals with physical or mental impairments. Across Europe, the design, coverage, and enforcement of disability support programs significantly influence whether individuals with health limitations can remain in or enter employment, providing a direct test of H5 (health and disability employment effects) and H6 (institutional moderation).

National disability support programs: Countries vary in the comprehensiveness of their disability policies. Social democratic regimes, including Sweden and Denmark, offer extensive wage subsidies, workplace accommodations, and rehabilitation programs. ESS Round 10 data indicate that individuals with disabilities in these countries have employment probabilities approximately 10–12 percentage points higher than their counterparts in less supportive regimes. Programs such as partial sick leave, flexible work arrangements, and vocational retraining enable individuals with moderate functional limitations to remain employed or re enter the workforce (OECD, 2020; Eurostat, 2021).

Conservative and Southern European regimes, in contrast, provide limited support. Disability benefits often substitute for employment rather than facilitate it, and workplace accommodations are rare. ESS data show that adults with disabilities in Italy, Spain, and Greece are 20–25 percentage points less likely to be employed than non disabled peers, highlighting persistent institutional barriers (Eurofound, 2017; OECD, 2018). The result is a significant labour market exclusion effect, especially among women and older workers, reflecting the compounded impact of gender, age, and health status on employment (H5, H6).

Policy effectiveness and labour market integration: Evidence from OECD and Eurostat evaluations suggests that active labour market interventions, including supported employment, vocational training, and targeted employer incentives, improve employment

outcomes for people with disabilities by 8–15 percentage points (OECD, 2018; European Commission, 2019). Cross country analyses confirm that integrating these measures with inclusive workplace policies such as anti discrimination enforcement and flexible scheduling substantially reduces employment penalties. ESS Round 10 supports these findings: countries with proactive disability policies show smaller gaps between disabled and non disabled individuals, particularly for women balancing care responsibilities (H5, H6).

Cross country institutional moderation: Disability employment outcomes are strongly conditioned by welfare regime type, labour market flexibility, and societal norms. Social democratic countries combine universal social protections with targeted supports, achieving higher inclusion. Liberal regimes, such as the UK and Ireland, display mixed outcomes, with benefits concentrated on active participants and less support for those with severe impairments. Conservative and Southern regimes often provide passive benefits with minimal employment facilitation, resulting in persistent labour market exclusion (Esping Andersen, 1990; Eurofound, 2017).

Research gaps: While prior research demonstrates that disability policies improve employment, few studies examine interaction effects between disability, gender, education, migration, and welfare regime in a unified framework. ESS Round 10, combined with OECD and Eurostat indicators, allows for robust multilevel analyses, linking individual health and disability characteristics to country level institutional supports, directly informing H5 and H6.

Summary: Disability support policies substantially influence employment outcomes for individuals with functional limitations. Cross country evidence shows that inclusive programs, workplace accommodations, and active labour market interventions mitigate employment penalties, while less supportive regimes perpetuate exclusion. Gender, education, and migration status interact with disability, reinforcing the importance of institutional context. ESS Round 10 provides rich individual level and cross country data, enabling empirical testing of H5 (health/disability employment effects) and H6 (institutional moderation), bridging theoretical predictions and policy relevant outcomes.

3.7 Institutional and Cross Country Studies

3.7.1 Comparative Employment Research

Cross country comparisons provide critical insights into how institutional structures, labour market policies, and welfare regimes shape employment outcomes. ESS Round 10, combined with OECD labour market indicators, enables systematic assessment of

variations in employment probabilities across 31 European countries, providing empirical grounding for H6 (institutional moderation of employment determinants).

Patterns of employment variation: Comparative research consistently finds that employment rates are highest in social democratic regimes, intermediate in liberal regimes, and lowest in conservative corporatist and Southern European countries (Esping Andersen, 1990; Thévenon, 2011). ESS data indicate that, for prime age adults, average employment probabilities range from 78–82% in Sweden and Denmark to 60–65% in Italy, Spain, and Greece, controlling for age, education, and health. These cross country differences suggest that institutional characteristics including active labour market programs, childcare support, and labour market flexibility mediate the translation of individual human capital into employment outcomes.

Institutional determinants: Comparative studies emphasize three key mechanisms:

1. Welfare state support – Social democratic regimes provide universal childcare, parental leave, and wage subsidies, facilitating higher female and immigrant employment (Gornick & Meyers, 2003; OECD, 2018). In Southern European countries, limited support for working parents reduces employment, particularly for women and recent immigrants.
2. Labour market regulation – Rigid hiring and firing rules, high employment protection legislation, and segmented labour markets constrain employment entry for non native, low skilled, or part time workers (Cappellari & Jenkins, 2004; Eurostat, 2021). Flexible labour markets in liberal regimes increase overall employment but may exacerbate inequalities for vulnerable groups.
3. Integration policies – Institutional support for immigrant labour market integration through language courses, credential recognition, and active employment programs reduces employment gaps between natives and immigrants, while lack of targeted support perpetuates disparities (H4, H6) (Dustmann et al., 2016; Kogan, 2011).

Cross country evidence from ESS and OECD: ESS Round 10 shows that the interaction between human capital, gender, family responsibilities, health, and migration status varies systematically by institutional context. For example, highly educated immigrant women in Nordic countries achieve employment probabilities similar to natives, whereas comparable individuals in Southern Europe face persistent gaps of 15–20 percentage points. OECD comparative studies corroborate these findings, demonstrating that institutional moderation accounts for a significant portion of variance in employment outcomes, even after controlling for individual level determinants (OECD, 2018).

Policy relevant insights: Comparative research underscores the importance of comprehensive, coordinated policy packages. Countries combining active labour market programs, family support policies, and inclusive integration measures achieve the best employment outcomes across gender, age, and migration status. In contrast, piecemeal approaches or minimal supports fail to fully mitigate structural inequalities, reinforcing the significance of H6.

Comparative cross country research highlights the central role of institutional and policy contexts in shaping employment outcomes. ESS and OECD evidence confirms that employment variations across Europe are not solely attributable to individual characteristics but are strongly mediated by welfare regimes, labour market flexibility, and integration policies. These insights provide a robust empirical foundation for testing H6, connecting individual determinants (human capital, gender, family, health, migration) to broader institutional frameworks, and offering guidance for effective policy interventions.

3.7.2 Welfare Regime Effects

European welfare regimes exert a profound influence on employment patterns by shaping the availability of social supports, labour market flexibility, and institutional incentives for workforce participation. ESS Round 10, combined with OECD and Eurostat data, allows for detailed cross country comparisons, revealing how social democratic, liberal, conservative, and Southern regimes differentially affect employment outcomes across gender, migration status, family structure, and health conditions.

Social democratic regimes: Countries such as Sweden, Denmark, and Norway provide universal childcare, extensive parental leave, and active labour market programs, creating highly inclusive labour markets. ESS analyses indicate that women in these countries exhibit the smallest employment gaps relative to men (2–4 percentage points), and immigrant employment gaps are reduced to 3–6 percentage points after controlling for education and experience (Gornick & Meyers, 2003; OECD, 2018). Disability and health related employment penalties are similarly mitigated through workplace accommodations and rehabilitation programs (Eurostat, 2021). These regimes demonstrate the interactive effect of welfare policies with human capital, labour supply, and institutional support, directly informing H2, H3, H4, H5, and H6.

Liberal regimes: Countries such as the UK and Ireland emphasize market based solutions with limited universal support. ESS data show higher overall employment rates than Southern Europe but larger disparities among vulnerable groups. For example, immigrant women and mothers face 10–15 percentage point employment deficits compared to natives, and disabled individuals experience persistent exclusion without targeted interventions (OECD, 2020). While education remains a strong predictor of employment (H1), policy gaps

limit the ability of these groups to fully translate human capital into labour market outcomes, highlighting H6.

Conservative corporatist regimes: Germany, Austria, and France feature employment protections and family benefits tied to traditional employment patterns, often benefiting men in standard full time jobs while constraining women and part time workers. ESS evidence indicates that motherhood penalties are higher, averaging 12–15 percentage points, and immigrant integration is slower due to limited policy supports outside standard employment trajectories (Esping Andersen, 1990; Cappellari & Jenkins, 2004). Health and disability penalties also persist unless individuals are employed in protected sectors, reinforcing H2, H3, H4, H5, and H6.

Southern European regimes: Italy, Spain, and Greece combine rigid labour markets with minimal family and integration support. ESS data reveal the largest employment gaps for women (15–20 percentage points), immigrants (12–18 points), and individuals with health limitations (20–25 points). High youth unemployment and limited childcare availability exacerbate barriers for highly educated women and immigrant workers, demonstrating the compounded effect of welfare design on labour market inclusion (Thévenon, 2011; Eurofound, 2017).

Cross regime comparisons: Integration of ESS, OECD, and Eurostat evidence demonstrates that policy design and welfare generosity moderate the effects of individual characteristics on employment. Social democratic regimes consistently outperform other regimes across gender, migration, and health dimensions, while Southern and conservative regimes show persistent inequalities. These patterns support the hypothesis that institutional context conditions the realization of human capital, labour supply, and integration mechanisms, providing empirical justification for H2–H6.

Welfare regime type is a decisive moderator of employment outcomes across Europe. Social democratic regimes enable inclusive labour markets that reduce gender, immigrant, and health related employment gaps, whereas liberal, conservative, and Southern regimes display persistent disparities. ESS Round 10, complemented by OECD and Eurostat indicators, allows for robust cross country analysis, linking institutional design to individual employment outcomes and supporting hypotheses H2–H6. Understanding these regime effects is critical for policy interventions aimed at enhancing labour market inclusivity across diverse populations.

3.8 Synthesis of Literature

This section integrates empirical findings across education, gender, family structure, immigration, health, and institutional contexts to provide a comprehensive overview of

European labour market determinants. By synthesizing cross country evidence, we identify systematic patterns, reinforce theoretical linkages to H1–H6, and highlight persistent research gaps that motivate the present study.

Education and human capital: Cross country evidence consistently demonstrates that higher educational attainment increases employment probabilities, confirming H1. ESS Round 10 data show that tertiary educated individuals are approximately 15–20 percentage points more likely to be employed than those with only secondary education, with variations by country reflecting differences in labour market flexibility and credential recognition (OECD, 2018; Eurostat, 2021). Notably, institutional factors moderate these effects, supporting H6: in Nordic countries, returns to education are fully realized across genders and for immigrants, whereas in Southern Europe, structural rigidity limits the employment benefits of high educational attainment.

Gender and employment: Evidence indicates persistent gender gaps despite rising female labour force participation. ESS analyses reveal that, on average, women face 10–15 percentage points lower employment probability than men, with the gap widening among mothers and immigrant women (H2, H3). Mechanisms include household responsibilities, part time labour prevalence, and differential access to childcare and parental leave (Gornick & Meyers, 2003; Blau & Kahn, 2013). Policy interventions such as subsidized childcare and flexible work arrangements substantially reduce these gaps, but effectiveness is highly regime dependent: social democratic regimes achieve near parity, whereas Southern European regimes maintain high penalties.

Family structure: Parenthood has gendered impacts, with mothers experiencing significant employment penalties (12–18 percentage points) and fathers sometimes receiving small premiums (2–5 points), particularly in conservative and liberal regimes (H2, H3). ESS and OECD data confirm that these effects interact with education and welfare policies. Highly educated mothers in Nordic countries face only minor employment reductions, reflecting both supportive childcare and labour flexibility, whereas in Southern Europe, penalties remain severe due to limited institutional support (Esping Andersen, 1990; Thévenon, 2011).

Immigration: Employment gaps between immigrants and natives remain substantial across Europe, particularly for recent arrivals and non EU migrants (H4). ESS data show average gaps of 12–18 percentage points, moderated by host country specific human capital and policy support. Integration programs, credential recognition, and active labour market interventions significantly reduce disparities in social democratic regimes, whereas limited support in Southern European countries exacerbates persistent gaps (H4, H6) (Dustmann et al., 2016; Kogan, 2011).

Health and disability: Health status is a strong predictor of employment. Poor physical or mental health reduces employment probabilities by 10–20 percentage points (H5), with effects amplified for women, older workers, and low educated individuals. ESS evidence demonstrates that social democratic regimes with inclusive disability programs and workplace accommodations mitigate these penalties, whereas Southern and conservative regimes maintain substantial employment gaps for individuals with disabilities (H5, H6) (OECD, 2018; Eurofound, 2017).

Institutional and welfare regime effects: Cross country evidence confirms that welfare regime type conditions the impact of all individual level determinants (H6). Social democratic regimes minimize gender, immigrant, and disability gaps; liberal regimes show moderate inclusion with higher variance; conservative and Southern regimes display persistent inequalities. ESS Round 10, combined with OECD and Eurostat indicators, underscores the interactive effects of human capital, labour supply constraints, discrimination, migration, and policy support on employment outcomes.

Integrated insights: Across these domains, several key patterns emerge:

1. Education consistently increases employment, but institutional support determines whether returns are fully realized (H1, H6).
2. Gender gaps persist, but are highly sensitive to childcare, parental leave, and labour market flexibility (H2, H3, H6).
3. Parenthood penalties are context dependent, varying by regime, gender, and education (H2, H3, H6).
4. Immigrant employment gaps reflect both human capital mismatches and policy deficiencies, with social democratic regimes offering the most effective mitigation (H4, H6).
5. Health and disability status strongly influence employment, but inclusive institutional supports can substantially reduce penalties (H5, H6).

This synthesis demonstrates that employment outcomes in Europe are shaped by the interplay of individual characteristics and institutional contexts. While education, gender, family, migration, and health independently affect employment probabilities, their effects are moderated by welfare regimes and national policies, confirming the need for multilevel, cross country analysis. ESS Round 10 provides the empirical foundation to address these gaps, enabling robust tests of H1–H6 and informing targeted policy interventions. This comprehensive understanding sets the stage for the following section, 3.9 Research Gaps,

which identifies areas underexplored in prior studies and justifies the proposed methodological approach in Chapter 4.

3.9 Research Gaps This Thesis Addresses

Despite extensive research on employment determinants, several critical gaps remain that limit both theoretical understanding and policy relevance. This thesis explicitly identifies these gaps and positions the ESS Round 10 dataset to address them, providing a strong empirical foundation for testing hypotheses H1–H6.

1. Cross country variation in education returns (H1, H6): While prior studies document that higher education increases employment probability (Blau & Kahn, 2013; OECD, 2018), few analyses systematically compare returns to education across multiple welfare regimes, accounting for interactions with gender, migration, and institutional factors. ESS Round 10 provides harmonized measures of educational attainment, enabling robust cross country analyses that quantify how institutional context moderates the effect of education on employment, addressing gaps left by single country or limited sample studies.

2. Persistent gender gaps under changing labour market conditions (H2, H3, H6): Research highlights ongoing female employment penalties, particularly related to parenthood and caregiving responsibilities (Gornick & Meyers, 2003; Thévenon, 2011). However, few studies integrate individual level labour supply constraints, household composition, and welfare regime characteristics simultaneously. Most existing literature either examines policy impacts in isolation or fails to account for cross country differences in childcare availability, parental leave duration, and labour market flexibility. By combining ESS individual data with country level indicators, this thesis can isolate how gender gaps are moderated by institutional structures, directly informing H2, H3, and H6.

3. Immigration related employment disparities (H4, H6): Immigrant employment gaps have been documented extensively (Dustmann et al., 2016; Kogan, 2011), yet there is limited evidence on how host country specific human capital and integration policies interact with individual attributes, such as education and gender, across multiple regimes. Many studies rely on administrative data from single countries, missing the comparative dimension. ESS Round 10 enables a cross national perspective, linking employment outcomes to duration of stay, recognition of foreign credentials, and policy supports, directly addressing H4 and H6.

4. Health and disability impacts (H5, H6): Although the negative relationship between poor health or disability and employment is well established (OECD, 2018; Eurofound, 2017), prior research often neglects interactions with

gender, migration, and institutional support. Existing studies frequently treat health effects in isolation, without considering how welfare regime characteristics, workplace accommodations, and social protections modify outcomes. ESS Round 10 provides rich health indicators, allowing multilevel analysis of health and disability penalties moderated by national institutions, testing hypotheses H5 and H6.

5. Integration of multiple determinants: A major limitation in current literature is the fragmented approach: education, gender, family, immigration, and health are often analyzed separately. There is limited research on their joint influence within a single framework, especially across diverse welfare regimes. By combining these factors and considering cross level institutional moderation, this thesis fills a crucial gap in understanding the complex, interactive determinants of employment, providing empirical tests for H1–H6 simultaneously.

6. Methodological limitations: Many previous studies rely on single level regression or country specific data, which may fail to capture endogeneity, selection bias, and multilevel interactions. ESS Round 10 enables advanced methodological approaches, including logistic regression with multilevel modelling, robustness checks, and potential integration of ML and deep learning techniques for predictive validation, ensuring that the analyses are both theoretically sound and empirically rigorous.

The existing literature highlights the determinants of employment but remains limited in cross country, multilevel, and integrated analyses. Specifically, gaps exist in understanding how education, gender, family structure, immigration, health, and institutional contexts interact to influence employment across Europe. ESS Round 10 provides the necessary individual level and country level data to address these limitations. This thesis leverages this unique dataset to test H1–H6, combining theoretical grounding, empirical rigor, and cross country comparability, ultimately generating novel insights with both academic and policy relevance.

3.10 Chapter Summary and Transition to Methodology

Chapter 3 provided a comprehensive review of empirical research on the determinants of employment in Europe, integrating evidence across education, gender, family structure, immigration, health, and institutional contexts. The analysis highlights systematic patterns, identifies gaps in the literature, and establishes the empirical foundation for the current study using ESS Round 10 data.

Education and human capital (H1, H6): Empirical studies confirm that higher educational attainment strongly predicts employment probability, yet returns are highly context dependent. Cross country comparisons reveal that individuals in social democratic regimes benefit most from their education, while those in Southern and conservative regimes face constraints due to rigid labour markets and limited policy support. These findings underscore the interactive effects of individual human capital and institutional structures, directly informing H1 and H6.

Gender and family structure (H2, H3, H6): Persistent gender gaps emerge across Europe, amplified by parenthood, caregiving responsibilities, and labour market segmentation. Social democratic regimes mitigate these disparities through extensive childcare, parental leave, and flexible work policies, whereas Southern and conservative regimes display higher penalties for mothers. Cross country evidence further demonstrates that family structure interacts with education and welfare policies, highlighting the importance of labour supply constraints and institutional context in shaping employment outcomes (H2, H3, H6).

Immigration and employment (H4, H6): Immigrants face persistent employment gaps relative to natives, with variation driven by host country specific human capital, integration policies, and welfare regimes. ESS and OECD evidence show that inclusive policies, credential recognition, and active labour market interventions substantially reduce these gaps, whereas restrictive or fragmented support maintains structural inequalities. These patterns emphasize the combined role of individual characteristics and institutions, providing empirical grounding for H4 and H6.

Health and disability (H5, H6): Poor health and disability remain significant barriers to employment, particularly for women, older adults, and low educated individuals. Social democratic regimes with inclusive workplace accommodations and active support programs mitigate employment penalties, while Southern and conservative regimes show persistent exclusion. This evidence demonstrates the moderating influence of welfare policies and institutional design on labour market outcomes, directly supporting H5 and H6.

Institutional and welfare regime effects (H2–H6): Comparative studies and ESS analyses illustrate that welfare regimes condition the magnitude and distribution of employment outcomes. Social democratic regimes achieve the highest inclusion across gender, immigration, and health dimensions, liberal regimes produce moderate outcomes, and conservative and Southern regimes display persistent inequalities. These findings confirm that institutional context is a key moderator, reinforcing the multilevel framework underpinning this thesis.

Critical analysis and synthesis:

While prior research provides valuable insights, several limitations persist: studies often focus on single countries, treat determinants in isolation, or lack harmonized cross country data. ESS Round 10 addresses these limitations by providing individual level measures across 31 countries, enabling the integration of education, gender, family, migration, and health variables with institutional indicators. This allows for multilevel, comparative analyses that capture both individual and structural determinants of employment, overcoming methodological and conceptual gaps in the literature.

Transition to methodology:

The gaps identified in Chapter 3 justify a robust empirical strategy integrating individual level data, country level institutional indicators, and advanced statistical methods. Chapter 4 will detail the ESS Round 10 dataset, variable construction, descriptive statistics, and the econometric and methodological framework employed to test H1–H6. The chapter will also outline the proposed use of logistic regression, multilevel models, and robustness checks, along with potential exploratory application of advanced machine learning techniques, ensuring both rigor and policy relevance.

Chapter 3 demonstrates that employment outcomes in Europe are shaped by the interplay of human capital, gender, family responsibilities, migration, health, and institutional context. The identified research gaps provide a clear rationale for the empirical strategy, directly linking theory to testable hypotheses. Chapter 4 builds on this foundation, describing the data, methodological choices, and analytical approach required to rigorously examine the determinants of employment across European countries.

Plan

“Chapter 4: Data and Methodology

This chapter describes the data, variable construction, and empirical strategy used to test Hypotheses H1–H6. It establishes the empirical validity of the analysis and clarifies how the theoretical framework is translated into testable econometric models.

4.1 Data Source: European Social Survey (ESS) Round 10

The analysis is based on ESS Round 10 (2020–2022), a cross-national, probability-based survey covering 31 European countries. The chapter outlines the ESS sampling design,

target population, and weighting strategy to ensure national representativeness. Country coverage is described and classified by welfare regime type, with brief justification for inclusion decisions. Data quality issues, including non-response and self-reported measures, are discussed.

4.2 Variables and Measurement

Employment status is defined as participation in paid work and coded consistently across countries. Independent variables include education (ISCED), gender, age, family structure (marital status and children), immigration status, and health. Country-level institutional indicators capture welfare regime characteristics. Control variables account for demographic and contextual factors. All measurements are theoretically motivated and explicitly linked to H1–H6.

4.3 Descriptive Statistics

This section presents sample characteristics and employment rates across countries and welfare regimes. Bivariate relationships between employment and key explanatory variables provide initial insights and motivate the multivariate analysis.

4.4 Econometric Methodology

Employment probabilities are estimated using logistic regression models, with results reported as average marginal effects. The estimation strategy follows a sequential modelling approach, gradually introducing individual, family, migration, health, and institutional variables. Interaction terms are used to test institutional moderation effects (H6). Potential sources of bias and endogeneity are discussed, and robustness is assessed using alternative specifications, including probit models and subsample analyses.

4.5 Analytical Plan

Each hypothesis (H1–H6) is mapped directly to specific model specifications and variables, ensuring transparent hypothesis testing and internal coherence.

4.6 Chapter Summary

The chapter concludes by reaffirming the suitability of the data and methods and prepares the transition to Chapter 5, where empirical results are presented.

Chapter 5: Empirical Results

This chapter presents the empirical findings of the study and evaluates Hypotheses H1–H6. It reports baseline results, explores subgroup and cross-country heterogeneity, and assesses the robustness of the estimates.

5.1 Introduction

The chapter briefly restates the empirical objectives and outlines the structure of the results section, emphasizing the progression from baseline models to more detailed subgroup and cross-country analyses.

5.2 Baseline Results: Full Sample

This section presents the main regression results for the full sample. Estimated effects of education, gender, family structure, immigration status, health, and institutional variables on employment probability are reported using average marginal effects. Each hypothesis (H1–H6) is explicitly tested and interpreted.

5.3 Subgroup Analyses

Results are disaggregated by key population groups to assess heterogeneity. Separate analyses are conducted by gender, age group, education level, and immigrant status. These models examine whether the magnitude and direction of effects differ across groups, with particular attention to gendered and migration-related employment gaps.

5.4 Cross-Country Heterogeneity

This section examines variation in employment effects across countries and welfare regimes. Country-specific estimates and regime-level comparisons are presented to assess institutional moderation. The analysis highlights systematic differences between social-democratic, liberal, conservative, and Southern welfare regimes.

5.5 Robustness Checks

The robustness of the findings is evaluated using alternative model specifications, including probit models, different variable codings, and sensitivity analyses. Consistency of results across specifications is discussed.

5.6 Chapter Summary

The chapter concludes by summarizing the main empirical patterns and identifying key findings that require further interpretation. It prepares the transition to Chapter 6, where the results are discussed in relation to theory, prior literature, and policy implications.

Chapter 6: Discussion

This chapter interprets the empirical findings in light of the theoretical framework and existing empirical literature. It evaluates support for Hypotheses H1–H6, explains unexpected results, and discusses the broader policy and research implications.

6.1 Summary of Main Findings

This section synthesizes the key empirical results across all models. It highlights central patterns related to education, gender, family structure, immigration status, health, and institutional context, with emphasis on cross-country variation and subgroup differences.

6.2 Interpretation of Results

The empirical findings are interpreted through the lenses of human capital theory, labour supply theory, discrimination theories, immigration economics, and institutional theory. Each hypothesis (H1–H6) is revisited and assessed based on the results. Particular attention is given to mechanisms explaining gender gaps, migration penalties, and institutional moderation effects, as well as to any results that diverge from theoretical expectations.

6.3 Comparison with Existing Literature

This section situates the findings within the broader empirical literature reviewed in Chapter 3. Areas of convergence and divergence with previous cross-country, gender, and migration studies are identified, and potential reasons for differences in results are discussed.

6.4 Policy Implications

Based on the empirical evidence, this section discusses implications for labour market and social policy. It focuses on education and training, childcare and parental leave, immigrant integration measures, health and disability support, and labour market regulation. Policy relevance is discussed in a comparative European context.

6.5 Limitations

This section critically reflects on the study's limitations, including data constraints, measurement issues, cross-sectional design, and modelling assumptions. The implications of these limitations for interpretation and generalizability are clearly acknowledged.

6.6 Directions for Future Research

The chapter concludes by identifying avenues for future research, including longitudinal extensions, causal identification strategies, richer institutional indicators, and potential applications of advanced modelling approaches.

Chapter 7: Conclusions

This chapter synthesizes the theoretical arguments, empirical findings, and policy implications of the study. It highlights the original contributions of the thesis, reflects on its broader relevance, and concludes the research in a coherent and forward-looking manner.

7.1 Main Contributions

This section summarises the core contributions of the thesis across three dimensions. The theoretical contribution lies in integrating human capital, labour supply, discrimination, immigration, and institutional theories into a unified framework for explaining employment outcomes. The empirical contribution derives from providing comparative evidence across European countries using harmonised survey data. The policy contribution relates to identifying institutional factors that shape employment inequalities.

7.2 Theoretical Implications

This section discusses how the findings refine existing theories. Human capital theory is revisited in light of heterogeneous returns to education across institutional contexts. Labour supply theory is reconsidered through evidence on gender and family constraints. Discrimination theories are informed by persistent employment gaps after controlling for observable characteristics. Institutional theory is strengthened by empirical support for welfare regime moderation effects.

7.3 Empirical Contributions

This section highlights the empirical value of the study. It documents systematic employment differences by education, gender, family status, migration background, and health across Europe. The analysis demonstrates how individual-level characteristics interact with country-level institutions, offering a more integrated understanding of labour market outcomes than single-country or single-factor studies.

7.4 Policy Relevance

This section translates the findings into policy-relevant insights. It discusses implications for education and training systems, childcare and parental leave policies, immigrant integration programmes, health and disability support, and labour market regulation. The emphasis is on evidence-based policy design tailored to different European institutional contexts.

7.5 Final Remarks

The chapter concludes by reflecting on the broader significance of the research for understanding employment inequality in Europe. It highlights the value of comparative approaches and integrated frameworks and offers a closing perspective on how future research and policy can build on the thesis's findings.

Integration with Project Workflow

Chapter 4: Links directly to /datasets and /notebooks (EDA, transformations, ML/ML validation).

Chapter 5: Outputs feed into /artifacts (models, regression results, pipelines).

Chapters 6–7: Discussion and conclusions interpret results while reflecting on methodology and policy implications.

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References

- Addo, F., & Chin, G. J. (2021). *The labor market integration of immigrant women in Europe: Context, theory, and evidence*. SN Social Sciences, 1(2), Article 279. <https://doi.org/10.1007/s43545-021-00279-3>
- Algan, Y., Dustmann, C., Gritz, A., & Manning, A. (2010). *The economic situation of first and second generation immigrants in Europe*. Journal of Ethnic and Migration Studies, 36(1), 1–30. <https://doi.org/10.1080/13691830903250862>
- Amuedo Dorantes, C., & de la Rica, S. (2007). *Labor market assimilation of immigrants in Spain*. Labour Economics, 14(6), 909–926.
- Angelov, N., Johansson, P., & Lindahl, E. (2016). *Parenthood and the gender gap in pay*. Journal of Labor Economics, 34(3), 545–579.
- Arrow, K. J. (1973). *The theory of discrimination*. In O. Ashenfelter & A. Rees (Eds.), *Discrimination in labour markets* (pp. 3–33). Princeton University Press.
- Barglowski, K., & Pustulka, P. (2018). Tightening early childcare choices: Gender and social class inequalities among Polish mothers in Germany and the UK. *Comparative Migration Studies*, 6, Article 36. <https://doi.org/10.1186/s40878-018-0102-6>
- Becker, G. S. (1964). *Human capital: A theoretical and empirical analysis, with special reference to education* (3rd ed.). Columbia University Press.
- Becker, G. S. (1981). *A treatise on the family*. Harvard University Press.
- Bernardi, F., Ballarino, G., & Pastore, F. (2011). Educational attainment and earnings assimilation of immigrants in Europe. *European Sociological Review*, 27(4), 542–560.
- Blau, F. D., & Kahn, L. M. (2013). Female labor supply: Why is the US falling behind? *American Economic Review*, 103(3), 251–256. <https://doi.org/10.1257/aer.103.3.251>
- Blau, F. D., & Kahn, L. M. (2017). The gender wage gap: Extent, trends, and explanations. *Journal of Economic Literature*, 55(3), 789–865. <https://doi.org/10.1257/jel.20160995>
- Blundell, R., Duncan, A., & Meghir, C. (2000). *Labour supply and intertemporal substitution*. Journal of Econometrics, 96(1), 89–121.
- Bol, T. (2015). *Returns to education in Europe: Common trends and persisting diversity*. Economic Policy, 30(81), 1–45.
- Budig, M. J., & England, P. (2001). *The wage penalty for motherhood*. American Sociological Review, 66(2), 204–225.
- Cappellari, L., & Jenkins, S. P. (2004). Modelling low income transitions. *Journal of Applied Econometrics*, 19(4), 593–610. <https://doi.org/10.1002/jae.762>
- Cappellari, L., & Jenkins, S. P. (2008). Estimating low pay transition models for Europe. *Economica*, 75(300), 255–279. <https://doi.org/10.1111/j.1468-0335.2007.00594.x>
- Dustmann, C., & Fabbri, F. (2003). *Language proficiency and labour market performance*. Economic Journal, 113(489), 695–717.
- Dustmann, C., Frattini, T., & Halls, C. (2016). *Assessing the labour market integration of migrants in Europe*. Economic Policy, 31(85), 497–550.
- European Foundation for the Improvement of Living and Working Conditions (Eurofound). (2020). *Living and working in Europe: European Quality of Life Survey*.

- European Social Survey (ESS). (2022). *Round 10 Data (2020–2022)*. ESS ERIC Data Archive. <https://www.europeansocialsurvey.org>
- European Institute for Gender Equality (EIGE). (2021). *Women and men in the EU: Facts and figures*.
- Esping Andersen, G. (1990). *The three worlds of welfare capitalism*. Princeton University Press.
- Ferre, Z., Triunfo, P., & Antón, J. I. (2023). Immigrant assimilation in health care utilisation in Spain. arXiv.
- Gornick, J. C., & Meyers, M. K. (2003). *Families that work: Policies for reconciling parenthood and employment*. Russell Sage Foundation.
- Haas, L., & Rostgaard, T. (2011). Fathers' workplace leave and gender equality. *Social Politics*, 18(1), 121–143.
- Husted, T. A., et al. (2000). Immigrant labour market performance in Europe. *Labour Economics*, 7(3), 607–629.
- Izquierdo, M. (2025). Reforms and in work poverty in the EU. *Economic Systems*. <https://doi.org/10.1016/j.ecosys.2025.101361>
- Juhyun, L. (2022). A comparative study of eight European countries: Life course events and female migrant labour market integration. *Journal of International and Comparative Social Policy*. Cambridge University Press.
- Kogan, I. (2011). *Selecting immigrants: Education, labour markets and wages*. (Book).
- Kotsadam, A., & Finseraas, H. (2011). The state intervenes in the battle of the sexes: Causal effects of paternity leave. *Social Science Research*, 40(6), 1611–1622.
- Kristen, C., Schmidt, R., & Mühlau, P. (2022). Educational selectivity and immigrants' labour market performance in Europe. *European Sociological Review*, 38(2), 252–268. <https://doi.org/10.1093/esr/jcab042>
- Leira, A. (1994). *Gender, state and family*. Cambridge University Press.
- Lundberg, S., & Pollak, R. A. (1993). Separate spheres bargaining and the marriage market. *Journal of Political Economy*, 101(6), 988–1010.
- Mandel, H., & Semyonov, M. (2006). A welfare state paradox: State interventions and women's employment opportunities. *American Journal of Sociology*, 111(6), 1910–1949.
- Marczuk, A. (2024). Literature review of comparative school to work research. *Journal for Labour Market Research*, 58, Article 19. <https://doi.org/10.1186/s12651-024-00375-w>
- Masselot, A. (2010). *Reconciling work and family life in EU law and policy*. Palgrave Macmillan.
- Matysiak, A., & Cukrowska Torzewska, E. (2021). *Gender and labour market outcomes*. In Research Handbook on the Sociology of the Family. Edward Elgar Publishing.
- Mincer, J. (1974). *Schooling, experience and earnings*. Columbia University Press.
- Molina, O., & Nowak, J. (2025). *Children and gender inequality: Evidence from cross national parental leave data*. Review of Economic Studies. (Forthcoming).
- OECD. (2018). *Education at a glance 2018: OECD indicators*. OECD Publishing. <https://doi.org/10.1787/eag-2018-en>
- OECD. (2019). *International Migration Outlook 2019*. OECD Publishing.
- OECD. (2020). *Employment Outlook 2020*. OECD Publishing.
- Pichler, F. (2011). The role of education in native–immigrant occupational attainment. *European Sociological Review*, 27(1), 111–129.

Ruhm, C. J. (1998). The economic consequences of parental leave mandates. *Journal of Health Economics*, 17(5), 531–555.

Schultz, T. W. (1961). Investment in human capital. *American Economic Review*, 51(1), 1–17.

Srinivasan, P., et al. (2022). *Labor market situation of refugees in Europe: Individual and contextual factors*. *Frontiers in Political Science*. <https://doi.org/10.3389/fpos.2022.977764>

Spence, M. (1973). Job market signaling. *Quarterly Journal of Economics*, 87(3), 355–374. <https://doi.org/10.2307/1882010>

Thévenon, O. (2011). *Family policies in OECD countries: A comparative analysis*. *Population and Development Review*, 37(1), 57–87.

Zimmert, F. (2023). Early child care and the employment potential of mothers: Evidence from difference in differences estimation. *Journal for Labour Market Research*, 57, Article 19. <https://doi.org/10.1186/s12651-023-00344-9>